

IT Support
Hierarchy

SOLVE > SUPPORT > DEPLOY

L1 L2 L3 SUPPORT

RIGHT ISSUE · RIGHT LEVEL · RIGHT EXPERTISE

Different
Levels
Same Goal!

L1

FIRST LINE SUPPORT

Quick Fixes, Happy Users

Handles common issues & basic support



Key Responsibilities

- ✓ Answer user queries (phone, email, chat)
- ✓ Resolve basic hardware & software issues
- ✓ Password resets & account access
- ✓ Troubleshoot common MS Office / Outlook
- ✓ Printer, peripheral & connectivity issues
- ✓ Log and escalate complex issues



Common Tools



Windows



Outlook



Teams



Ticketing
System



Example Issues

- PC not turning on
- Email not working
- Printer offline
- Password reset request
- Software installation

L2

SECOND LINE SUPPORT

Deeper Analysis, Smarter Solutions

Handles complex issues & system-level problems



Key Responsibilities

- ✓ Advanced troubleshooting & diagnostics
- ✓ System configuration & optimization
- ✓ Network connectivity & VPN issues
- ✓ Application support (ERP, CRM, etc.)
- ✓ Active Directory & user management
- ✓ Coordinate with third-party vendors



Common Tools



Windows



Active
Directory



Microsoft 365



Network
Tools



Example Issues

- Domain login issues
- Exchange / email configuration
- VPN / remote access problems
- Application errors
- Network / LAN connectivity

L3

THIRD LINE SUPPORT

Expertise. Escalation. Resolution.

Handles critical, complex & rare issues



Key Responsibilities

- ✓ Deep system, network & security analysis
- ✓ Root cause investigation
- ✓ Advanced configuration & scripting
- ✓ Resolve complex infrastructure issues
- ✓ Work with vendors & product teams
- ✓ Create permanent fixes & documentation



Common Tools



Cisco



VMware



Wireshark



PowerShell



Example Issues

- Network outages & routing issues
- Active Directory / Group Policy failures
- Firewall / security incidents
- Performance & scalability problems
- System crashes / kernel issues

Small
Problems
Big Impact



Solve
Build
Improve



Zafran Khan
IT Support Specialist

Better Support → Stronger Systems → A Connected Future



Karachi, Pakistan

L3 IT Support / System Administrator - Quick Cheat Sheet (V1)

Real World Issues | Troubleshooting | Solutions | Best Practices

Stay Calm.
Think Logical.
Solve It 😊

1. Active Directory & User Management

Problem:

- User can't login / access resources
- Account locked / disabled
- Group policy not applying
- Computer not joining domain
- Permission denied



Solution:

- ✓ Check account status (ADUC)
- ✓ Reset password / unlock account
- ✓ Verify group policy and OU
- ✓ Check DNS & domain join settings
- ✓ Check permissions (GPO / NTFS)

2. Microsoft 365 / Exchange Online

Problem:

- Email not sending / receiving
- Outlook not working
- User can't access OneDrive/SharePoint
- License issue
- Mailbox quota exceeded



Solution:

- ✓ Check mailbox & license status
- ✓ Verify mail flow / rules
- ✓ Restart Outlook / clear cache
- ✓ Check service health (M365 admin)
- ✓ Manage storage / remove old data

3. Network Connectivity

Problem:

- No internet / slow internet
- Intermittent connection
- VPN not connecting
- DNS not resolving
- Website not loading



Solution:

- ✓ Check cable / Wi-Fi / link light
- ✓ Verify IP, gateway, DNS (ipconfig)
- ✓ Test ping / tracer
- ✓ Check firewall / proxy settings
- ✓ Restart router / modem / device

4. Windows Server & Services

Problem:

- Server not reachable
- Service not starting
- High CPU / memory usage
- Disk space full
- Backup failure



Solution:

- ✓ Check network & power
- ✓ Review Event Viewer logs
- ✓ Restart service / server
- ✓ Check resource usage
- ✓ Verify backup & storage

5. Hardware Failures

Problem:

- PC not turning on
- No display
- Overheating
- Random shutdown
- Peripheral not working



Solution:

- ✓ Check power / cables / PSU
- ✓ Reseat RAM / GPU
- ✓ Test with known good parts
- ✓ Check temperature / fan
- ✓ Update BIOS / firmware

6. Printer & Peripheral Issues

Problem:

- Printer not printing
- Paper jam
- Scanner not working
- USB device not detected
- Driver not found



Solution:

- ✓ Check power / connection
- ✓ Verify driver & port
- ✓ Check print queue / spooler
- ✓ Test with another cable/port
- ✓ Reinstall/update driver

7. Security & Access

Problem:

- Suspicious login attempt
- Malware / virus alert
- Firewall blocking access
- Account compromise
- System infected



Solution:

- ✓ Check logs & alerts
- ✓ Scan with antivirus/EDR
- ✓ Review firewall rules
- ✓ Reset credentials
- ✓ Follow incident response plan

8. Backup & Disaster Recovery

Problem:

- Backup job failed
- Restore not working
- Missing files
- Storage full
- RPO/RT0 not met



Solution:

- ✓ Check backup logs
- ✓ Verify storage availability
- ✓ Test restore process
- ✓ Clean old backups
- ✓ Review backup policy & schedule

9. System Updates & Patching

Problem:

- Updates not installing
- Reboot required repeatedly
- Patch conflicts
- Software outdated
- Security vulnerability



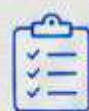
Solution:

- ✓ Check Windows Update / WSUS
- ✓ Review event logs
- ✓ Install updates manually if needed
- ✓ Check for software compatibility
- ✓ Schedule maintenance window

10. Documentation & Best Practices

Problem:

- No proper documentation
- Recurring issues
- Lack of standard process
- Asset tracking missing
- Silo resolution time



Solution:

- ✓ Maintain SOPs / runbooks
- ✓ Document changes & fixes
- ✓ Keep asset register updated
- ✓ Use ticketing system
- ✓ Follow ITIL best practices

Good documentation saves time and prevents repeat issues.



BASIC IT TROUBLESHOOTING

WINDOWS MACHINE

Quick solutions for common Windows issues

01 NO INTERNET CONNECTION



- Check for Wi-Fi Signal Range
- Confirm Network Configuration
- Run commands in cmd prompt
ipconfig/release
ipconfig/renew
ipconfig/flushdns
- Reset modem/Laptop

02 LAPTOP SLOW



- Open Task Manager (Ctrl+ Shift+ Esc) & disable startup apps if required
- Run disk cleanup or use storage sense
- Scan for malware in your laptop with windows defender

03 LAPTOP BSOD



- Make sure you note the error code
- Boot your laptop in safe mode
- Update device drivers
- Run sfc /scannow

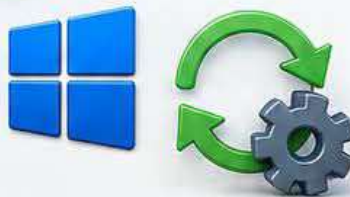
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04 APP CRASHING



- Open Task Manager & End the task related to that APP.
- Delete the APP & re-install
- Check for any windows update
- Check for any compatibility mode

05 WINDOWS UPDATE ISSUE



- Try to use troubleshooter
- Clean update cache
net stop wuauclt
net stop bits
del C:\windows\software
\distribution\download /s /q
net start wuauclt
net start bits

06 WINDOWS LOGGING ISSUE



- Reset Local account password
- Try to boot in a Safe mode
- Check for any windows update
- For domain users, please check network & domain controller availability



PRO TIP

Keep your system
healthy & productive.



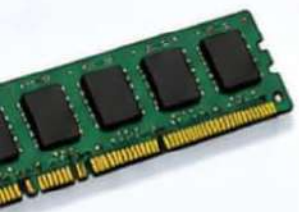
KEEP UPDATED



STAY SECURE



BACKUP REGULARLY



HOW TO CHECK RAM HEALTH

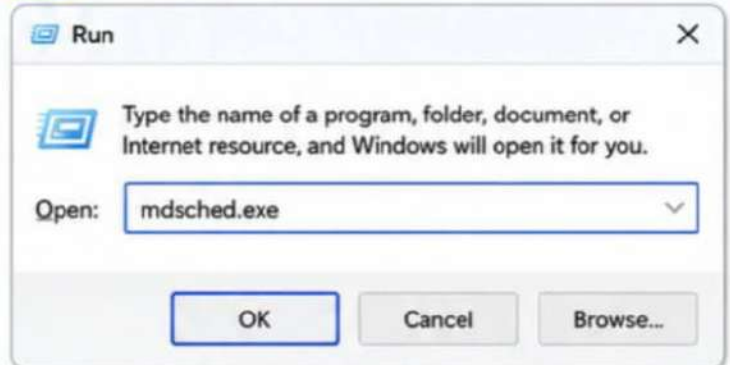
METHOD 1: WINDOWS MEMORY DIAGNOSTIC (BUILT-IN)

Windows has a built-in tool to check your RAM for problems.

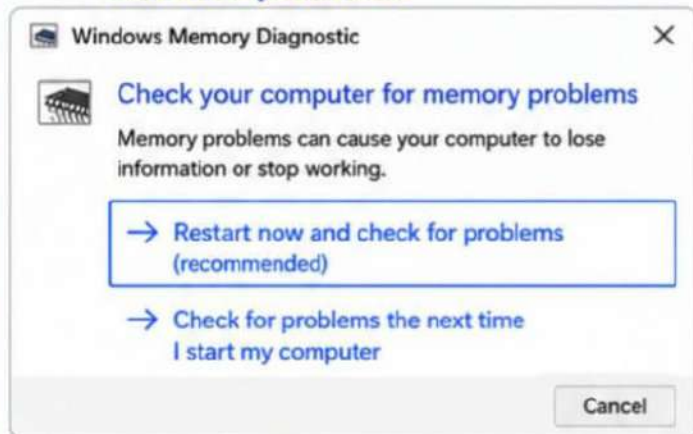
1 Press Win + R.



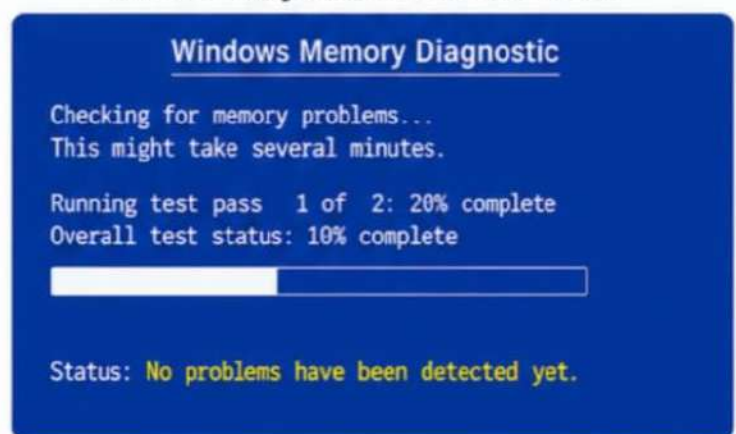
2 Type `mdsched.exe` and press Enter.



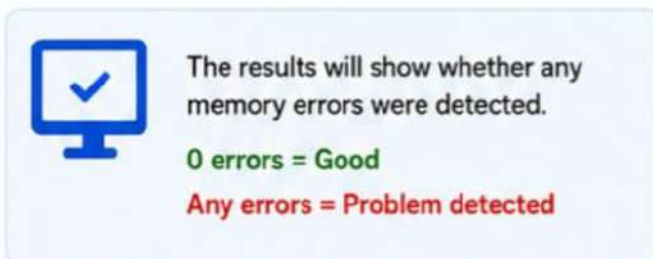
3 Select **Restart now and check for problems**.



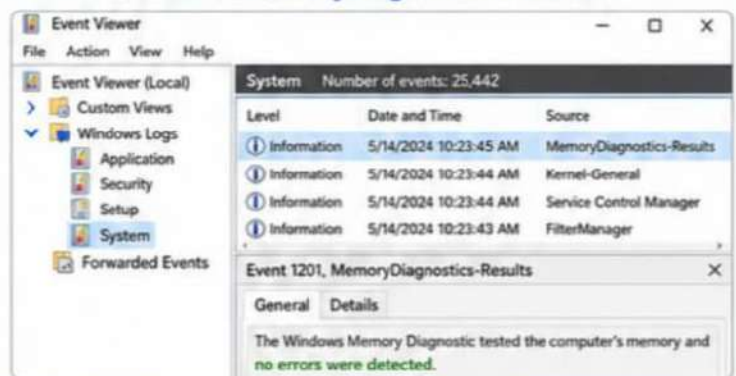
4 Your PC will restart and test the RAM. The test may take several minutes.



5 After Windows starts, view the results in Event Viewer.

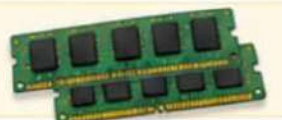


6 Press Win + X → **Event Viewer** Go to **Windows Logs** → **System** Search for **MemoryDiagnostics-Results**.



TIP

If problems are found, try removing and reseating the RAM sticks one by one to identify the faulty module, or replace the defective RAM.



BACKUP WINDOWS 11 COMPLETE IMAGE TO PENDRIVE

BACKUP TODAY, RESTORE TOMORROW – STAY SAFE!



**WINDOWS
11 PRO**



LAPTOP STORAGE
512 GB SSD
(Only OS)



PENDRIVE
32 GB
(For Boot/Recovery)



BACKUP DRIVE
128 GB+
(For Full Image)



IMPORTANT:

32 GB pendrive is NOT enough for full image.

Use it only for boot/recovery. Use 128 GB+ drive for the backup.



32 GB



128 GB+

BOOT USB

BACKUP DRIVE

1 GET THE BACKUP DRIVE

Use a 128 GB or larger
USB drive or external SSD.

Make sure it is empty or
you can format it.



2 DOWNLOAD RESCUEZILLA

Go to the official website and
download the latest ISO.

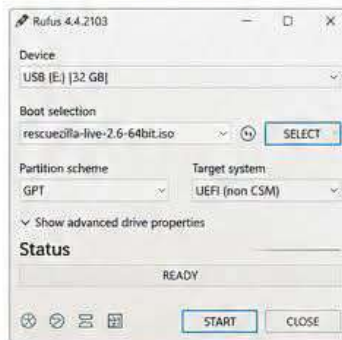


<https://rescuezilla.com>

3 MAKE 32 GB PENDRIVE BOOTABLE (WITH RUFUS)

1. Insert your 32 GB pendrive.
2. Open Rufus.
3. Select your pendrive under Device.
4. Select the Rescuezilla .iso file.
5. Leave default settings.
6. Click Start.
7. Confirm it can be erased.
8. Wait until it finishes.

Now your pendrive is a
Rescuezilla Boot USB.



4 CONNECT BACKUP DRIVE

Connect your 128 GB+ USB drive /
external SSD to the laptop.



5 BOOT FROM 32 GB PENDRIVE

1. Shut down the laptop.
2. Insert the Rescuezilla 32 GB pendrive.
3. Turn on the laptop.
4. Open the Boot Menu
(Common keys: F12, F9, F11, Esc, F2).
5. Select the USB drive.



6 CREATE THE IMAGE (BACKUP)

In Rescuezilla:



Choose
Backup



Select your
internal 512 GB
SSD



Select your
external backup
drive



Choose a
location &
Start

Let it finish. Do not turn off the laptop during the backup.

7 RESTORE WINDOWS (WHEN NEEDED)

1. Connect the backup drive with your image.
2. Insert the 32 GB Rescuezilla USB.
3. Boot from the USB.
4. Select Restore.
5. Select your saved backup image.
6. Select the laptop's internal SSD as destination.
7. Start the restoration.
8. When finished, remove USB drives and restart.



DONE!

Your laptop will be restored to
the same state – Windows,
programs, settings, drivers,
and files included.



TIPS

- Use a reliable USB drive or external SSD.
- Keep your backup drive safe.
- You can create a new image whenever
you want an updated backup.



IMPORTANT

Always back up before major
changes, updates, or installing
new software.



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How to Reset Windows 11

Step-by-Step Guide

Resetting Windows 11 can fix system issues, improve performance and give your PC a **fresh start**.



1 Open Settings



Press
Win + **I**
to open **Settings**.

2 Go to System



In Settings, click on **System** from the left sidebar.

3 Open Recovery



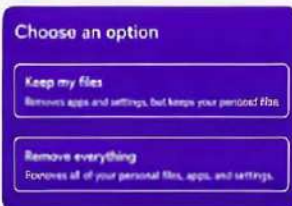
Scroll down and click on **Recovery**.

4 Click Reset PC



Under Recovery options, click **Reset PC**.

5 Choose an Option



Select one option:

- **Keep my files** (Recommended)
Removes apps and settings, but keeps your personal files.
- **Remove everything** (Full reset)
Removes all of your personal files, apps, and settings.

6 Additional Settings



Click **Change settings** to customize your reset preferences (optional).

7 Confirm Your Choice



Review the information and click **Reset** to continue.

8 Wait for the Process



Windows will reset your PC. This may take some time. Please wait...

9 PC Will Restart



Your PC will restart several times and **Windows 11** will be reset. **All done!** ✓



Back up your important files before resetting.



Ensure your PC is plugged in during the process.



Have your Windows product key ready.



The process may take 20-60 minutes depending on your PC.



Your PC will be like new, with better speed and stability!

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www.techwalejha.com

WHICH STEP WAS MOST HELPFUL FOR YOU?
COMMENT BELOW! 💬



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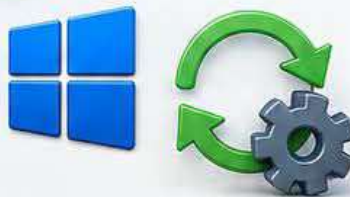
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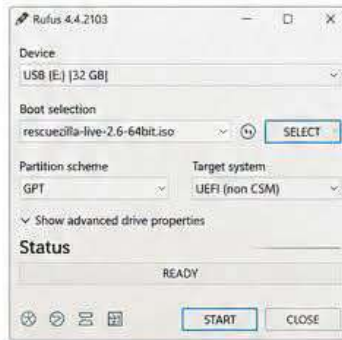


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HOW TO SHARE A PRINTER ON WINDOWS

Share your printer so other computers on the same network can use it.



1 CONNECT THE PRINTER

- Connect the printer to your PC via USB or Wi-Fi.
- Make sure it is installed and prints correctly.



2 ENABLE NETWORK DISCOVERY

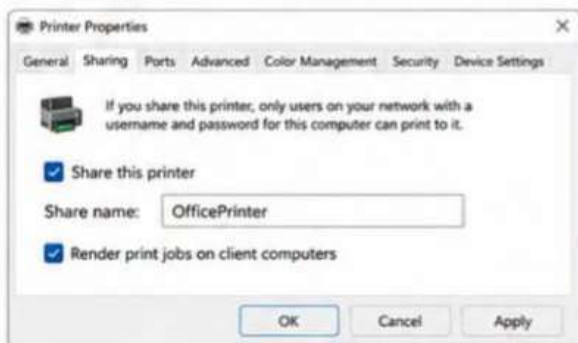
1. Open Control Panel > Network and Sharing Center.
2. Click Change advanced sharing settings.
3. Turn on:

- ☒ Network Discovery
- ☒ File and Printer Sharing



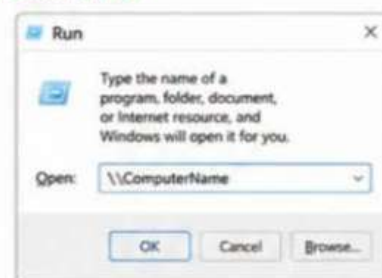
3 SHARE THE PRINTER

1. Open Settings > Bluetooth & devices > Printers & scanners.
2. Select your printer.
3. Click Printer properties.
4. Open the Sharing tab.
5. Check Share this printer.
6. Give it a simple share name (e.g., OfficePrinter).
7. Click Apply > OK.



4 CONNECT FROM ANOTHER COMPUTER

1. Press Win + R, type:
`\\ComputerName`
or
`\\IP_Address`
2. Press Enter.
3. Double-click the shared printer.
4. Windows will install the required drivers automatically (or prompt you if needed).



ALTERNATIVE METHOD



Go to:
Settings > Bluetooth & devices > Printers & scanners > Add device
Choose Add manually if the shared printer isn't found automatically.



Windows Settings



Bluetooth & devices



Printers & scanners



Add device
(Add manually)

TIPS



Both computers should be on the same network.



Keep the host PC powered on while others use the shared printer.



Allow File and Printer Sharing through Windows Defender Firewall if the printer isn't detected.



Make sure you have the necessary permissions to share and connect the printer.

WINDOWS STARTUP REPAIR

FIX BOOT PROBLEMS

Startup Repair helps fix problems that keep Windows from starting correctly.

WHAT IT CAN FIX



Corrupted boot files



Windows failing to start



Endless restart loops



Boot configuration problems



Black screen during startup



Problems caused by failed updates or system changes

HOW TO RUN STARTUP REPAIR

1



Start the PC and wait for Windows Recovery Environment.

2



Select **Advanced options**.

3



Go to **Troubleshoot** → **Advanced options**.

4



Select **Startup Repair**.

5



Choose your Windows account.

6



Enter your password if prompted.

7



Let Windows diagnose and repair the startup problem.

8



Restart the PC and check whether Windows boots normally.

IF STARTUP REPAIR DOESN'T WORK



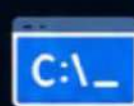
Uninstall Updates



System Restore



Startup Settings → Safe Mode



Use Command Prompt for advanced repairs



Important: Startup Repair is designed to repair boot-related problems; it normally **doesn't delete** your personal files.



WINDOWS SERVER BACKUP

Protect your data. Recover with confidence.



WHAT IS WINDOWS SERVER BACKUP?

Windows Server Backup is a built-in Microsoft feature that helps you back up your server and recover your data in case of hardware failure, corruption, or accidental deletion.



INSTALL WINDOWS SERVER BACKUP



Using Server Manager

- 1 Open Server Manager.
- 2 Select Manage → Add Roles and Features.
- 3 Click Next until you reach Features.
- 4 Check Windows Server Backup.
- 5 Click Install.



Using PowerShell

```
Install-WindowsFeature Windows-Server-Backup
```

CREATE A BACKUP

- 1 Open Windows Server Backup from the Start menu.
- 2 Click Local Backup.
- 3 Select Backup Schedule (for recurring backups) or Backup Once.
- 4 Choose:
 - Full server
 - Custom (specific files/volumes)
- 5 Select the backup destination:
 - Local disk
 - External USB drive
 - Network shared folder
- 6 Start the backup.



RESTORE DATA

- 1 Open Windows Server Backup.
- 2 Select Recover.
- 3 Choose the backup location.
- 4 Select the backup date.
- 5 Choose what to recover:



Files and Folders



Volumes



Applications



System State



- 6 Complete the recovery wizard.

KEY FEATURES



Full Server Backup

Backs up all critical volumes and installed applications.



System State Backup

Includes registry, boot files, Active Directory (if applicable), COM+ database, and other system components.



Volume Backup

Backs up entire disks or partitions.



File and Folder Backup

Protects selected files and folders.



Bare Metal Recovery (BMR)

Restores a server from scratch after a hardware failure.

POWERSHELL BACKUP EXAMPLES

Backup the C: drive to E:

```
wbadmin start backup -backupTarget:E: -include:C: -quiet
```

Backup the entire server:

```
wbadmin start backup -backupTarget:E: -allCritical -quiet
```

Backup system state:

```
wbadmin start systemstatebackup -backupTarget:E: -quiet
```

COMMON wbadmin COMMANDS

COMMAND	DESCRIPTION
<code>wbadmin get versions</code>	Lists available backups
<code>wbadmin get disks</code>	Displays available disks
<code>wbadmin start backup</code>	Starts a backup
<code>wbadmin start recovery</code>	Recovers files or volumes
<code>wbadmin start systemstatebackup</code>	Creates a system state backup
<code>wbadmin start systemstaterecovery</code>	Restores the system state

BEST PRACTICES



Schedule automatic backups (daily or more frequently for critical servers).



Keep backups on a separate physical disk, network share, or dedicated backup storage.



Follow the 3-2-1 backup rule: 3 copies of your data, on 2 different media, with 1 copy stored offsite.



Regularly test backup restoration to ensure backups are usable.



Monitor backup jobs and available storage space.



Encrypt backup storage when it contains sensitive data.



Windows Server Backup is ideal for small to medium environments.

For advanced capabilities like centralized management, deduplication-aware backups, cloud integration, or application-aware backups across many servers, consider enterprise backup solutions alongside or instead of the built-in tool.



NETWORKING ABBREVIATIONS



LAN

Local Area Network



WAN

Wide Area Network



MAN

Metropolitan Area Network



PAN

Personal Area Network



WLAN

Wireless Local Area Network



VPN

Virtual Private Network



IP

Internet Protocol



TCP

Transmission Control Protocol



UDP

User Datagram Protocol



HTTP

HyperText Transfer Protocol



HTTPS

HyperText Transfer Protocol Secure



FTP

File Transfer Protocol



SFTP

Secure File Transfer Protocol



SMTP

Simple Mail Transfer Protocol



POP3

Post Office Protocol



IMAP

Internet Message Access Protocol



DNS

Domain Name System



DHCP

Dynamic Host Configuration Protocol



MAC

Media Access Control



NIC

Network Interface Card



ISP

Internet Service Provider



OSI

Open Systems Interconnection



ARP

Address Resolution Protocol



ICMP

Internet Control Message Protocol



SSID

Service Set Identifier



NAT

Network Address Translation



VLAN

Virtual Local Area Network



VoIP

Voice over Internet Protocol



ACL

Access Control List



BGP

Border Gateway Protocol



RIP

Routing Information Protocol



OSPF

Open Shortest Path First



SNMP

Simple Network Management Protocol



SSH

Secure Shell



RDP

Remote Desktop Protocol



RDP

Remote Desktop Protocol



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EASY TO LEARN

Quick reference to key networking abbreviations.



IMPROVE SKILLS

Understand protocols and technologies faster.



BOOST EFFICIENCY

Save time with this handy networking cheat sheet.



BUILD CONFIDENCE

Perfect for students, professionals, and IT enthusiasts.



STAY AHEAD

Essential abbreviations for a stronger networking foundation.

Windows 11

Slow Performance Fix



Speed up your PC and improve performance with these simple and effective solutions!



1



DISABLE UNNECESSARY STARTUP APPS

Settings → Apps → Startup
Disable apps that you don't need when Windows starts.

2



CHECK BACKGROUND PROCESSES

Press Ctrl + Shift + Esc → Open Task Manager
Close apps using high CPU / Memory / Disk.

3



FREE UP STORAGE SPACE

Go to: Settings → System → Storage

- Remove temporary files
- Enable Storage Sense

4



UPDATE WINDOWS & DRIVERS

- Install latest Windows updates
- Update Graphics, Chipset, and Storage drivers

5



REDUCE VISUAL EFFECTS

Press Win + R → Type: sysdm.cpl
Advanced → Performance → Settings
Select Adjust for best performance.

6



SCAN FOR MALWARE

Run: Windows Security →
Virus & threat protection → Full Scan

7



OPTIMIZE DRIVE

Search: Defragment and Optimize Drives

- Optimize HDD
- SSD should use TRIM optimization

8



UPGRADE HARDWARE (IF NEEDED)

- ✓ Add more RAM
- ✓ Replace HDD with SSD
- ✓ Keep at least 15–20% free storage

CHECK SYSTEM PERFORMANCE

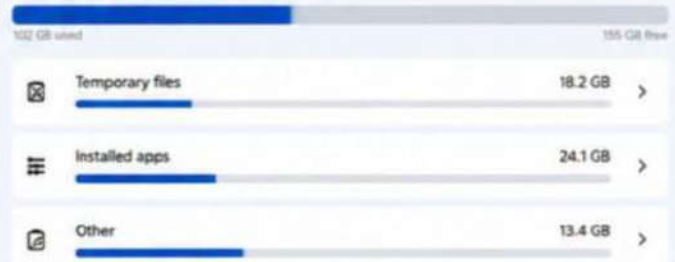
Press Ctrl + Shift + Esc → Task Manager



STORAGE CLEANUP

Settings > System > Storage

Local Disk (C:) - 237 GB



Show more categories



A clean startup + SSD + enough RAM can dramatically improve Windows 11 speed.

USEFUL COMMANDS (RUN AS ADMINISTRATOR)



SYSTEM FILE CHECKER

```
sfc /scannow
```

Scans and repairs corrupted system files.



REPAIR WINDOWS IMAGE

```
DISM /Online /Cleanup-Image /RestoreHealth
```

Repairs Windows image and system health.



Keep your Windows 11 updated, clean and optimized for the best performance!



LAN CABLE

TWO LIGHT BLINKING



GREEN LIGHT (Link / Activity)

-  **Solid Green** | Connection Established
-  **Blinking Green** | Data Activity
-  **Off** | No Link Detected



ORANGE LIGHT (Speed / Duplex)

-  **Solid Orange** | Gigabit Connection
-  **Blinking Orange** | Network Activity
-  **Off** | 10/100 Mbps Connection

SUMMARY



GREEN LIGHT

Shows Link & Activity Status

-  **Solid Green** : Connection Established
-  **Blinking Green** : Data Activity
-  **Off** : No Link Detected



ORANGE LIGHT

Shows Connection Speed

-  **Solid Orange** : Gigabit Connection
-  **Blinking Orange** : Network Activity
-  **Off** : 10/100 Mbps Connection



Reliable Connection



High Performance



Smooth Network



Stay Connected

WINDOWS BOOT LOOP PROBLEM FIX

Stuck in a restart loop? Here's how to fix Windows Boot Loop step by step.



COMMON CAUSES



Corrupted system files



Faulty updates or drivers



Hard drive / SSD errors



Incorrect BIOS settings



Third-party software conflicts



QUICK TIPS



Backup important data regularly.



Keep Windows and drivers updated.



Check disk health periodically.

STEP-BY-STEP FIX

1



Force Restart 2-3 Times

Turn on your PC and as soon as Windows starts, force shutdown. Repeat 2-3 times until "Preparing Automatic Repair" appears.

2



Enter Windows Recovery Environment (WinRE)

After the automatic repair screen, click Advanced options > Troubleshoot > Advanced options.

3



Run Startup Repair

Go to Troubleshoot > Advanced options > Startup Repair. Let Windows fix startup problems automatically.

4



Use Command Prompt (If Startup Repair Fails)

In Advanced options > Command Prompt, run:

```
sfc /scannow  
DISM /Online /Cleanup-Image /RestoreHealth  
chkdsk C: /f /r
```

Restart your PC after each command.

5



System Restore

Go to Advanced options > System Restore and choose a restore point before the issue started.

6



Uninstall Recent Updates/Drivers

Go to Advanced options > Uninstall Updates or Startup Settings > Safe Mode, then remove recent updates or drivers.

7



Last Option: Reset or Reinstall Windows

If nothing works, go to Advanced options > Reset this PC or perform a clean installation.



STILL NOT WORKING?

The issue might be due to hardware failure (HDD/SSD/RAM). Check your hardware or consult a professional technician.



BIOS NOT DETECTING HARD DRIVE

BIOS can't see your hard drive?
Here's how to fix it!



If your hard drive is not showing in BIOS, your system won't boot from it.

COMMON CAUSES



Loose or faulty
SATA / Power cable



Hard drive failure



BIOS settings issue



Faulty SATA port



Outdated BIOS

ADDITIONAL TIPS

- ✓ Remove all external devices and check.
- ✓ If using M.2/NVMe SSD, make sure it is seated properly.
- ✓ If drive is making unusual sounds, it may be physically damaged.



CHECK & SOLUTIONS

1



CHECK CABLE CONNECTIONS

Turn off PC. Reconnect SATA data cable and power cable properly.

2



TRY ANOTHER CABLE

Use another SATA data cable and power cable.

3



CHECK ANOTHER SATA PORT

Connect the drive to a different SATA port on the motherboard.

4



CHECK BIOS SETTINGS

Enter BIOS Setup → Check if SATA controller is enabled and set to AHCI mode.

5



TEST THE HARD DRIVE

Connect the drive to another PC or use an external enclosure to check if it works.

6



UPDATE BIOS

Update to the latest BIOS version from the motherboard or laptop manufacturer.



STILL NOT DETECTING?

The hard drive may be faulty.
Replace the drive if needed.



Always backup your data regularly to avoid data loss.

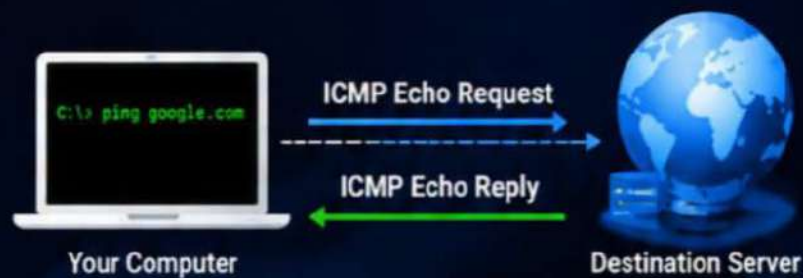


If you're not comfortable, seek help from a professional technician.

PING COMMAND

Check Network Connectivity & Response Time

Ping is a network troubleshooting command used to check whether a device is **reachable** over a network and measure the **response time** between two devices.



BASIC PING COMMAND

```
C:\> ping google.com
```

Checks whether Google's server is reachable and shows response time.

Example Output

```
Reply from 142.250.195.14: bytes=32 time=20ms TTL=117
Reply from 142.250.195.14: bytes=32 time=21ms TTL=117
Reply from 142.250.195.14: bytes=32 time=19ms TTL=117
Reply from 142.250.195.14: bytes=32 time=20ms TTL=117
```

Ping statistics for 142.250.195.14:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 19ms, Maximum = 21ms, Average = 20ms

- bytes=32 → Data packet size sent
- time=20ms → Round-trip response time
- TTL=117 → Time To Live value (network hops limit)

COMMON PING COMMANDS IN WINDOWS

1		ping 192.168.1.1 Ping an IP Address	Checks connectivity to a router or network device.
2		ping 8.8.8.8 -t Continuous Ping	Keeps sending ping requests until stopped. Stop using: Ctrl + C
3		ping google.com -n 5 Ping with Specific Number of Requests	Sends only 5 ping packets.
4		ping google.com -l 1000 Ping with Larger Packet Size	Sends a 1000-byte packet. Useful for testing packet handling.
5		ping google.com -4 Check IPv4 Only	Forces ping to use IPv4.
6		ping google.com -6 Check IPv6 Only	Forces ping to use IPv6.

PING RESULTS MEANING

- Reply from...**
Device is reachable.
- Request timed out**
Device not responding, Firewall blocking, or Network issue.
- Destination host unreachable**
No route to destination or Gateway problem.
- Ping transmit failed**
Network adapter or TCP/IP issue.

PING TROUBLESHOOTING EXAMPLES

- Check your router **ping 192.168.1.1** → Tests connection to your router or gateway.
- Check internet connectivity **ping 8.8.8.8** → Tests connectivity to a reliable public IP.
- Check DNS resolution **ping google.com** → If IP ping works but domain ping fails → DNS issue.

💡 If IP ping works but domain ping fails → DNS issue.

PING COMMAND QUICK SUMMARY

Command	Purpose	Example
ping IP or Domain	Test device connectivity	ping 192.168.1.1 / ping google.com
ping -t	Continuous monitoring	ping 8.8.8.8 -t
ping -n <number>	Set number of packets	ping google.com -n 5
ping -l <size>	Change packet size (in bytes)	ping google.com -l 1000
ping -4	Test IPv4	ping google.com -4
ping -6	Test IPv6	ping google.com -6



TIPS

Ping is one of the first commands IT professionals use to diagnose network problems.



Simple, Fast & Powerful!

WINDOWS BOOT CONFIGURATION DATA (BCD)

EXPLAINED

Boot Configuration Data (BCD) is a database used by Windows to control how the operating system starts. It replaced the old **BOOT.INI** file used in older Windows versions.



WHAT DOES BCD STORE?

- Windows boot loader entries
- Installed operating systems
- Boot menu options
- Default operating system selection
- Safe Mode settings
- Debugging options
- Boot timeout settings

WHERE IS BCD LOCATED?

UEFI SYSTEMS:

EFI\Microsoft\Boot\BCD

LEGACY BIOS SYSTEMS:

\Boot\BCD

The BCD file is usually stored in the **System Reserved** partition or **EFI System Partition (ESP)**.

HOW WINDOWS USES BCD DURING STARTUP

- 1 PC powers on
↓
- 2 BIOS / UEFI starts
↓
- 3 Windows Boot Manager loads
↓
- 4 Boot Manager reads the BCD store
↓
- 5 Windows Boot Loader starts Windows using BCD settings

COMMON BCD PROBLEMS

- "The Boot Configuration Data file is missing"
- "0xc000000f" Boot Configuration Data error"
- Windows stuck in boot loop
- Dual boot menu not appearing

USEFUL BCD REPAIR COMMANDS

Open Command Prompt as Administrator

- | | | | |
|------------------------------------|--|---|--|
| View BCD entries
bcdedit | Scan Windows installations
bootrec /scanos | Rebuild BCD
bootrec /rebuildbcd | Repair boot files
bcdboot C:\Windows |
|------------------------------------|--|---|--|



TIP: Never manually edit the BCD file. Use **BCDEdit** (**bcdedit**) or recovery tools because incorrect changes can prevent Windows from booting.



REMEMBER

Keep your system updated and create a system restore point regularly. It can save you from boot problems!



KNOWLEDGE IS POWER

Understand BCD, troubleshoot smartly, and keep your Windows running!



HOW TO CREATE A VENTOY BOOTABLE USB

Create a multiboot USB drive in just a few steps!

1

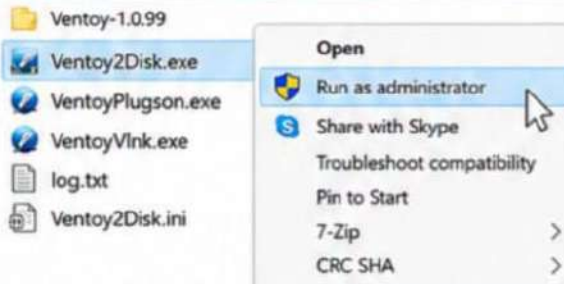


1

DOWNLOAD AND EXTRACT VENTOY

Download Ventoy from the official website and extract the ZIP file.

2



2

RUN VENTOY2DISK.EXE AS ADMINISTRATOR

Right-click on Ventoy2Disk.exe and select "Run as administrator".

3

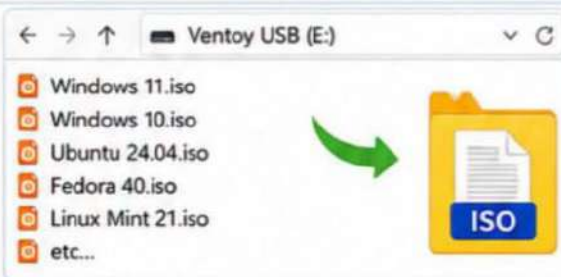


3

SELECT THE CORRECT USB DRIVE AND CLICK INSTALL

Choose your USB drive carefully and click "Install" to begin.

4



4

COPY YOUR ISO FILES TO THE USB

Copy your ISO files directly to the USB drive. No need to reformat for every ISO!



TIP

With Ventoy, you can keep multiple Windows, Linux, and other bootable ISO files on one USB drive.



IMPORTANT: Installing Ventoy will erase the selected USB drive. Back up your files first.





ULTIMATE WINDOWS RAM TOOLKIT

OPTIMIZE • CLEAN • MONITOR • SPEED UP



A POWERFUL ALL-IN-ONE TOOL TO MONITOR, CLEAN AND OPTIMIZE YOUR PC'S MEMORY

HOW TO USE



WHAT IS IT?



Ultimate RAM Toolkit is a Batch script tool for Windows that provides 17 essential utilities to monitor, clean, repair and optimize your system's memory and performance.

KEY FEATURES



RAM MONITORING

View RAM status, live usage and detailed memory information.



SYSTEM MANAGEMENT

Open Task Manager, Resource Monitor, Startup folder and more.



CLEAN & OPTIMIZE

Clear clipboard, temp files, standby cache and restart Explorer.



DIAGNOSTICS

Run Windows Memory Diagnostics to check RAM problems.



PERFORMANCE BOOST

Optimize settings and manage heavy background tasks.

IMPORTANT NOTE

This tool performs safe system maintenance. It does NOT magically increase physical RAM. Windows manages memory automatically.

17 POWERFUL TOOLS IN ONE

- View RAM Status
- Full System Memory Info
- Open Resource Monitor
- Open Task Manager
- Clear Clipboard Memory
- Restart Windows Explorer
- Close Frozen Apps
- Open Startup Folder
- Windows Memory Diagnostics
- RAM Usage Live Monitor
- Clear Standby RAM Cache
- Empty Temporary Files
- Show Heavy Background Tasks
- Open Advanced System Settings
- Open Performance Options
- FULL RAM OPTIMIZATION MODE**
- Exit

★ ONE CLICK, MANY BENEFITS

CLEANER SYSTEM • MORE FREE RAM • BETTER PERFORMANCE

PARTIAL VIEW OF THE SCRIPT

```
Ultimate_RAM_Toolkit.bat

@echo off
setlocal EnableExtensions
title ULTIMATE WINDOWS RAM TOOLKIT
color 0A

:: =====
:: ULTIMATE WINDOWS RAM TOOLKIT
:: Windows RAM monitoring and
:: maintenance utility
:: =====

:: Check for Administrator privileges
net session >nul 2>&1
if %errorlevel% neq 0 (
    echo Requesting Administrator
    privileges ...
    powershell -NoProfile -ExecutionPolicy
    Bypass -Command "Start-Process -FilePath\
    '%~f0' -Verb RunAs"
    exit /b
)

:MENU
cls
echo.
echo =====
echo ULTIMATE WINDOWS RAM TOOLKIT
echo =====
echo.
echo 1. View RAM Status
echo 2. Full System Memory Info
echo 3. Open Resource Monitor
echo 4. Open Task Manager
echo ... (remaining menu items)
echo.
set /p "choice=Select an option: "

if "%choice%"=="1" goto RAMSTATUS
if "%choice%"=="2" goto MEMORYINFO
if "%choice%"=="3" goto RESOURCE
...
```



FULL SCRIPT CONTAINS ALL
17 FUNCTIONS AND LOGIC

WHAT YOU GET



FASTER PC

Frees up memory and system resources



MORE STABLE

Closes frozen apps and fixes minor issues



BETTER PERFORMANCE

Optimized settings for smooth experience



CLEANER SYSTEM

Removes junk & temporary files



EASY TO USE

Simple menu interface for everyone



PRO TIP

Run the "FULL RAM OPTIMIZATION MODE" regularly to keep your PC in top shape!

Windows 11

Slow Performance Fix



Speed up your PC and improve performance with these simple and effective solutions!



1



DISABLE UNNECESSARY STARTUP APPS

Settings → Apps → Startup
Disable apps that you don't need when Windows starts.

2



CHECK BACKGROUND PROCESSES

Press Ctrl + Shift + Esc → Open Task Manager
Close apps using high CPU / Memory / Disk.

3



FREE UP STORAGE SPACE

Go to: Settings → System → Storage

- Remove temporary files
- Enable Storage Sense

4



UPDATE WINDOWS & DRIVERS

- Install latest Windows updates
- Update Graphics, Chipset, and Storage drivers

5



REDUCE VISUAL EFFECTS

Press Win + R → Type: sysdm.cpl
Advanced → Performance → Settings
Select Adjust for best performance.

6



SCAN FOR MALWARE

Run: Windows Security →
Virus & threat protection → Full Scan

7



OPTIMIZE DRIVE

Search: Defragment and Optimize Drives

- Optimize HDD
- SSD should use TRIM optimization

8



UPGRADE HARDWARE (IF NEEDED)

- ✓ Add more RAM
- ✓ Replace HDD with SSD
- ✓ Keep at least 15–20% free storage

CHECK SYSTEM PERFORMANCE

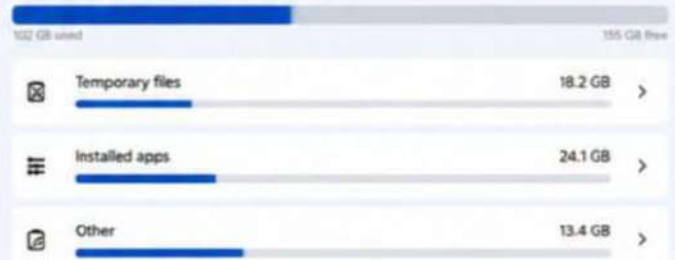
Press Ctrl + Shift + Esc → Task Manager



STORAGE CLEANUP

Settings > System > Storage

Local Disk (C:) - 237 GB



A clean startup + SSD + enough RAM can dramatically improve Windows 11 speed.

USEFUL COMMANDS (RUN AS ADMINISTRATOR)



SYSTEM FILE CHECKER

```
sfc /scannow
```

Scans and repairs corrupted system files.



REPAIR WINDOWS IMAGE

```
DISM /Online /Cleanup-Image /RestoreHealth
```

Repairs Windows image and system health.



Keep your Windows 11 updated, clean and optimized for the best performance!



Wi-Fi Driver vs Wi-Fi Adapter



What's the Difference?



Wi-Fi ADAPTER	Wi-Fi DRIVER
 HARDWARE (Physical Device)	 SOFTWARE (Program)
 Connects your PC to Wi-Fi	 Helps Windows communicate with the adapter
 Can be built-in inside your laptop or a USB adapter	 Installed in Windows
 Examples: Intel Wi-Fi card, USB Wi-Fi dongle	 Examples: Intel Wireless driver, Realtek Wi-Fi driver
 Can be physically replaced	 Can be updated, rolled back, or reinstalled



SIMPLE EXAMPLE



Wi-Fi Adapter

= The hardware



Wi-Fi Driver

= The translator



Your adapter receives the wireless signal, while the driver allows Windows to properly communicate with and control that hardware.



COMMON PROBLEM

If the Wi-Fi adapter is detected but Wi-Fi isn't working, the driver may be missing, corrupted, outdated, or incompatible.



CHECK HERE

Device Manager → Network adapters is a good place to check both.



REMEMBER: You can have a perfectly working Wi-Fi adapter, but without the correct driver, Windows may not be able to use it properly.

BLUE DUMP ERROR **FIXED**

Troubleshoot and fix Blue Screen (Blue Dump) errors in Windows.

BEFORE



Your device ran into a problem and needs to restart.
We're just collecting some error info, and then we'll restart for you.

20% complete



For more information about this issue and possible fixes, visit
<https://www.windows.com/stopcode>

If you call a support person, give them this info:
Stop code: DRIVER_IRQL_NOT_LESS_OR_EQUAL
What failed: myfault.sys



AFTER



Restarting

 **Blue Dump Error Fixed Successfully!**

COMMON CAUSES



Faulty or
Incompatible
Drivers



Memory
(RAM)
Issues



Corrupted or
Failing Hard
Drive / SSD



Windows
System File
Corruption



Overheating
Hardware

HOW TO FIX



Update or
Reinstall
Drivers



Run Windows
Memory
Diagnostic



Check Disk
(chkdsk)
for Errors



Repair System
Files (SFC /
DISM Scan)



Clean System
& Ensure
Proper Cooling

QUICK FIX STEPS

1



Safe Mode
Boot into Safe Mode
to isolate the issue.

2



Update Drivers
Update all drivers,
especially display
and storage drivers.

3



Run SFC & DISM
Open CMD (Admin)
and run:
sfc /scannow
DISM /Online /Cleanup-
Image /RestoreHealth

4



Check Disk
Run chkdsk /f /r
to fix file system
errors.

5



Test RAM
Use Windows Memory
Diagnostic to check
for memory issues.

6



Restart
Restart your PC and
check if the Blue
Dump error is fixed.



PRO TIP: Keep Windows and drivers updated and maintain your system regularly to prevent Blue Screen errors.



RESULT: System stable and running smoothly without Blue Dump errors.



ULTRA-ADVANCED FIREWALL TECHNICIAN TOOLKIT



20 POWERFUL COMMANDS FOR WINDOWS FIREWALL

Run Command Prompt as **Administrator** >_



- Show all firewall profiles**

```
netsh advfirewall show allprofiles
```

Displays the status of Domain, Private, and Public profiles.
- Show current firewall settings**

```
netsh advfirewall show currentprofile
```

Shows settings of the current active profile.
- Enable all firewall profiles**

```
netsh advfirewall set allprofiles state on
```

Enables firewall protection for all profiles.
- Disable all profiles**

```
netsh advfirewall set allprofiles state off
```

Disables firewall for all profiles. (Troubleshooting only)
- Reset firewall configuration**

```
netsh advfirewall reset
```

Resets all firewall settings to default.
- Display every firewall rule**

```
netsh advfirewall firewall show rule name=all
```

Displays all inbound and outbound rules.
- Search rules by name**

```
netsh advfirewall firewall show rule name="Remote Desktop"
```

Find and display rules by specific name.
- Block inbound TCP port 445**

```
netsh advfirewall firewall add rule name="Block SMB 445" dir=in action=block protocol=TCP localport=445
```

Blocks inbound traffic on TCP port 445.
- Block outbound TCP port 445**

```
netsh advfirewall firewall add rule name="Block SMB Outbound 445" dir=out action=block protocol=TCP remoteport=445
```

Blocks outbound traffic on TCP port 445.
- Allow inbound HTTPS**

```
netsh advfirewall firewall add rule name="Allow HTTPS" dir=in action=allow protocol=TCP localport=443
```

Allows inbound traffic on TCP port 443 (HTTPS).
- Block a specific application**

```
netsh advfirewall firewall add rule name="Block Application" dir=out action=block program="C:\Path\App.exe" enable=yes
```

Blocks a specific application from accessing the network.
- Delete an application rule**

```
netsh advfirewall firewall delete rule name="Block Application"
```

Deletes the specified application rule.
- Allow traffic from a trusted IP**

```
netsh advfirewall firewall add rule name="Trusted IP" dir=in action=allow remoteip=192.168.1.50
```

Allows inbound traffic from a trusted IP address.
- Block a suspicious IP**

```
netsh advfirewall firewall add rule name="Block Suspicious IP" dir=in action=block remoteip=192.168.1.100
```

Blocks inbound traffic from a specific IP address.
- Enable firewall logging (dropped)**

```
netsh advfirewall set allprofiles logging droppedconnections enable
```

Enables logging for dropped/blocked connections.
- Enable logging of successful connections**

```
netsh advfirewall set allprofiles logging allowedconnections enable
```

Logs all successful (allowed) connections.
- Display firewall log location**

```
netsh advfirewall show allprofiles | findstr /i "LogFileName"
```

Displays the location of the firewall log file.
- Export firewall configuration**

```
netsh advfirewall export "C:\Firewall_Backup.wfw"
```

Exports current firewall configuration to a file for backup.
- Import saved firewall configuration**

```
netsh advfirewall import "C:\Firewall_Backup.wfw"
```

Imports firewall configuration from a saved file.
- Open advanced firewall console**

```
wf.msc
```

Opens Windows Firewall with Advanced Security.



TECHNICIAN SAFETY TIPS

- ✓ Always export (backup) firewall configuration before making changes.
- ✓ Be careful with RDP (3389), SMB (445), broad IP rules, and disabling firewall.
- ✓ Incorrect rules can lock you out or expose the network.
- ✓ Test your rules and monitor logs after applying changes.
- ✓ Document rules for future reference.

BEFORE MAKING MAJOR CHANGES, BACKUP FIRST!

```
netsh advfirewall export "C:\Firewall_Backup.wfw"
```



BACKUP



INSPECT



CREATE NARROW RULE



TEST



MONITOR LOGS



DOCUMENT



REMOVE TEMP RULES



SECURE TODAY, PROTECT TOMORROW!

STRONGER NETWORK

SAFER SYSTEMS

SMARTER TECHNICIANS





ACTIVE DIRECTORY TUTORIAL

LAB ENVIRONMENT (RECAP)



DEVICE	ROLE	IP ADDRESS
DC01	Windows Server 2022 (Domain Controller)	192.168.10.10 (Static)
CLIENT01	Windows 11 Pro (Client)	DHCP
DC02 (Optional)	Windows Server 2022 (Additional DC)	192.168.10.11 (Static)

STEP 3

INSTALL AD DS ROLE

(Active Directory Domain Services)

In this step, we will install the Active Directory Domain Services (AD DS) role on our server (DC01). This role provides the core AD functionality.



PREREQUISITES

- ✓ Windows Server installed on DC01
- ✓ Static IP address configured
- ✓ Server name: DC01
- ✓ Internet connection (for updates)
- ✓ Logged in as Administrator

WHY AD DS ROLE?

AD DS stores information about users, computers, groups and other resources and makes it available to authorized users and administrators in the network.



ABOUT AD DS ROLE

Active Directory Domain Services (AD DS) is the core component of Active Directory. It handles:

- ✓ User & Group Management
- ✓ Authentication & Authorization
- ✓ Policies (via Group Policy)
- ✓ Directory Database (NTDS.dit)
- ✓ Replication
- ✓ Global Catalog

STEP-BY-STEP INSTALLATION PROCESS

1 Open Server Manager

Click on the **Server Manager** icon on the taskbar and open the dashboard.



2 Add Roles and Features

Click on "Manage" in the top-right corner and select "Add Roles and Features".



3 Select Installation Type

Choose "Role-based or feature-based installation" and click Next.



4 Select Destination Server

Select "Select a server from the server pool" and choose DC01, then click Next.



5 Select Server Role

Select "Active Directory Domain Services" from the roles list and click Next.



6 Confirm & Install

Click **Next** on the features page, review the summary, and click **Install**.



WHAT HAPPENS NEXT?



✓ After installation, we will promote this server to a Domain Controller in the next step.

VERIFICATION

After installation is complete, verify that the role is installed. In Server Manager, you will see AD DS in the dashboard.



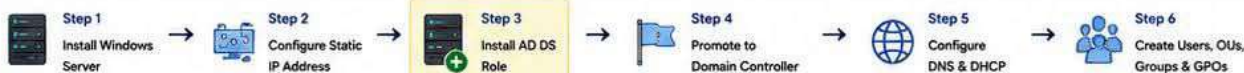
IMPORTANT NOTES

- AD DS cannot be removed after a Domain Controller is promoted.
- Ensure proper DNS and Static IP before installing AD DS.
- A clean and updated server ensures a smooth installation.
- Use strong administrator passwords.



A stable foundation today builds a secure Active Directory tomorrow.

UPCOMING STEPS



ACTIVE DIRECTORY TUTORIAL

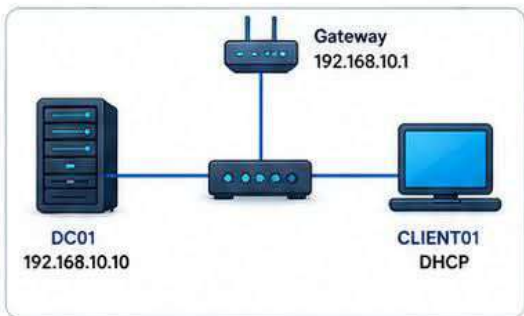
STEP 2

CONFIGURE STATIC IP ADDRESS

FOR DOMAIN CONTROLLER (DC01)



A static IP address is essential for a Domain Controller so that other computers can always find and communicate with it reliably.



LAB ENVIRONMENT			
DEVICE	ROLE	IP ADDRESS	NOTES
DC01	Windows Server 2022 (Domain Controller)	192.168.10.10 (Static)	Primary DNS & DC
CLIENT01	Windows 11 Pro (Client)	DHCP	Joins Domain later
DC02 (Optional)	Windows Server 2022 (Additional DC)	192.168.10.11 (Static)	For Replication

PREREQUISITES

- ✓ Windows Server installed on DC01
- ✓ Network adapter connected
- ✓ You have access as Administrator
- ✓ Know your network details (IP, Subnet, Gateway, DNS)

EXAMPLE NETWORK DETAILS

IP Address	: 192.168.10.10
Subnet Mask	: 255.255.255.0
Default Gateway	: 192.168.10.1
Preferred DNS	: 192.168.10.10

STEP-BY-STEP CONFIGURATION

1 Open Network Connections

Click on the network icon in the taskbar and open **Network & Internet Settings**.

2 Change Adapter Options

Click on **Change adapter options**.

3 Select Network Adapter

Right-click the active network adapter and select **Properties**.

4 Select IPv4

In the Properties window, select **Internet Protocol Version 4 (TCP/IPv4)** and click **Properties**.

5 Use Static IP Address

Select **Use the following IP address** and enter the details.

6 Apply Settings

Click **OK** to close all windows and apply the settings.

VERIFY CONFIGURATION

Open **Command Prompt** and run the following commands:

```
ipconfig /all
```

Check that the IP address, Subnet Mask, Gateway and DNS are set correctly.

```
ping 192.168.10.1
```

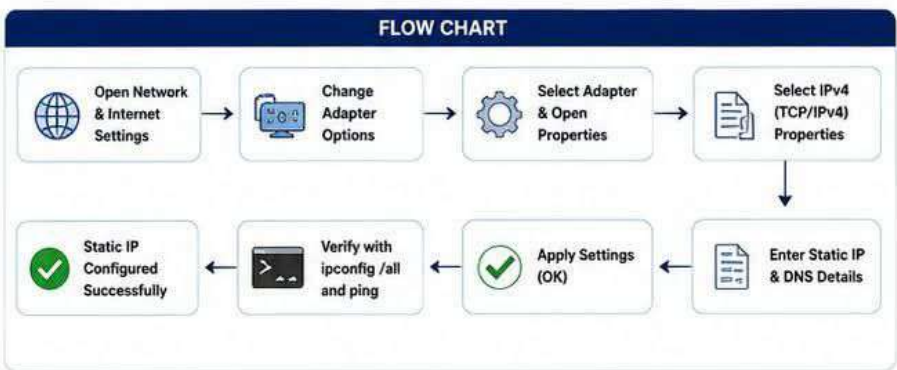
Should reply successfully (tests gateway).

```
ping DC01
```

Should reply successfully (tests local server name resolution via DNS).

Example Output (ipconfig /all)

```
Ethernet adapter Ethernet:  
Connection-specific DNS Suffix . : corp.example.com  
IPv4 Address. . . . . : 192.168.10.10  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : 192.168.10.1  
DNS Servers . . . . . : 192.168.10.10  
NetBIOS over Tcpip. . . . . : Enabled
```



IMPORTANT NOTES

- Always assign a static IP to Domain Controllers.
- Preferred DNS should be the DC's own IP address.
- All clients in the domain should use the DC as their DNS server.
- If IP changes, update DNS records and reconfigure clients.
- Avoid using 127.x.x.x or public DNS (like 8.8.8.8) for DC.

BEST PRACTICES

- Use a reserved/static IP from your DHCP scope.
- Document all IP settings.
- Ensure proper DNS resolution.





DRIVER PROBLEMS & DRIVER UPDATES

✓ Keep your PC running smooth, stable & secure



COMMON DRIVER PROBLEMS



MISSING DRIVER

Device not detected or shows unknown device.



OUTDATED DRIVER

Poor performance, errors or features not working.



CORRUPTED DRIVER

Causes crashes, Blue Screen or device not working.



INCOMPATIBLE DRIVER

After update, device stops working or system unstable.

HOW TO UPDATE DRIVERS (EASY STEPS)

- 1 Open Device Manager (Press Windows + X)
- 2 Right-click on the device
- 3 Select "Update driver"
- 4 Choose "Search automatically for drivers"
- 5 Follow instructions and restart PC



IMPORTANT DRIVERS TO KEEP UPDATED



Graphics Driver



Network / Wi-Fi Driver



Chipset Driver



Audio Driver



Bluetooth Driver



Storage Driver

BEST PRACTICES

- ✓ Create a restore point before updating.
- ✓ Download drivers from official websites only.
- ✓ Avoid third-party driver updater tools.
- ✓ Update only when needed or when facing issues.



TIP

Regular driver updates keep your system stable, secure and performing at its best.

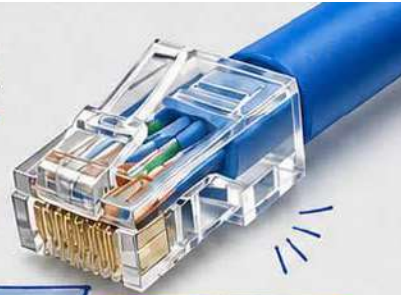


Let's Connect!

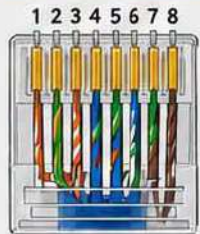
RJ45 Crimping Guide

T568B Wiring Standard

FOR NETWORKING • CCTV • IP CAMERAS • NVR • SWITCHES • ROUTERS



T568B PINOUT (8P8C)



RJ45 Plug
(Clip Away)

- | | | |
|---|--|----------------|
| 1 | | White / Orange |
| 2 | | Orange |
| 3 | | White / Green |
| 4 | | Blue |
| 5 | | White / Blue |
| 6 | | Green |
| 7 | | White / Brown |
| 8 | | Brown |

TOOLS YOU'LL NEED



RJ45
Connector
(8P8C)



Cat5e / Cat6
Cable



RJ45
Crimping Tool



Cable
Stripper



LAN Cable
Tester

STEP-BY-STEP CRIMPING PROCESS

1



Strip about
25 mm (1 inch)
of the outer jacket.

2



Untwist
the wire pairs.

3



Arrange
the wires in
T568B order.

4



Straighten and
trim the wires
evenly.

5



Insert the wires
fully into the
RJ45 connector.

6



Crimp the connector
firmly using
the tool.

7



Check the connector
and test the cable
before use.

TEST THE CABLE



CORRECT RESULT

Main: 1 2 3 4 5 6 7 8
Remote: 1 2 3 4 5 6 7 8 ✓

FAULT EXAMPLES

Open : 1 2 3 X 5 7 8
Short : 1 2 3 3 6 7 8
Cross : 1 2 4 3 6 7 8

Always test the cable to ensure proper connectivity.

IMPORTANT TIPS

- ✓ Always use T568B on both ends for straight-through cable.
- ✓ Use good quality cable (Cat5e or Cat6) and connectors.
- ✓ Make sure all wires reach the front of the connector.
- ✓ Trim wires evenly before inserting.
- ✓ Crimp firmly for good contact.
- ✓ Test every cable before installation.



COMMON APPLICATIONS



IP CAMERA
TO SWITCH / POE



NVR
TO SWITCH



SWITCH
TO SWITCH



ROUTER
TO MODEM / SWITCH



PC / COMPUTER
TO NETWORK



GOOD CONNECTION

• STABLE NETWORK

• BETTER PERFORMANCE



IPMI / iLO / iDRAC

REMOTE SERVER MANAGEMENT

Remote server management allows administrators to monitor, control, and troubleshoot servers remotely even when the operating system is down or the server is not booting.



IPMI

Intelligent Platform Management Interface

Industry-standard remote management protocol that works through a dedicated management port.

- ✓ Remote power ON/OFF/Restart
- ✓ Hardware health monitoring
- ✓ Temperature & fan status
- ✓ Event logging & alerts
- ✓ Console Redirection



HPE iLO

Integrated Lights-Out

HPE server remote management technology built into HPE ProLiant servers.

- ✓ Remote BIOS configuration
- ✓ Virtual media (ISO mounting)
- ✓ Remote console access
- ✓ Firmware updates
- ✓ Server health monitoring



Dell iDRAC

Integrated Dell Remote Access Controller

Dell PowerEdge server management solution for remote monitoring, control and automation.

- ✓ Remote server deployment
- ✓ Hardware monitoring
- ✓ Crash recovery
- ✓ Virtual console
- ✓ Automated alerts



QUICK COMPARISON

Feature	IPMI	HPE iLO	Dell iDRAC
Vendor	Multi-vendor Standard	HPE Servers	Dell Servers
Remote Power Control	✓	✓	✓
Remote Console Access	✓	✓	✓
Virtual Media (ISO/USB)	✓	✓	✓
Hardware Monitoring	✓	✓	✓
Advanced Automation	Limited	High	High

WHY USE IT?



Manage servers from anywhere



Troubleshoot without physical access



Install OS remotely



Reduce downtime



Monitor hardware failures early

USED IN



Data Centers



Enterprise Servers



Cloud Infrastructure

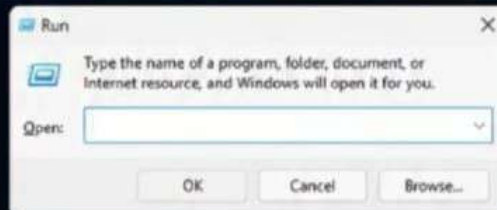


IT Operations

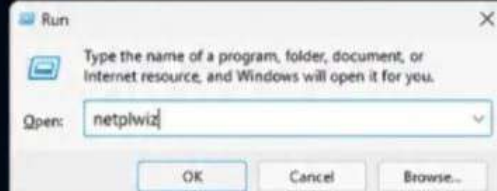
SECRETLY USING YOUR PC?

CHECK THIS HIDDEN WINDOWS FEATURE BEFORE IT'S TOO LATE!

1 Press **Windows + R**.



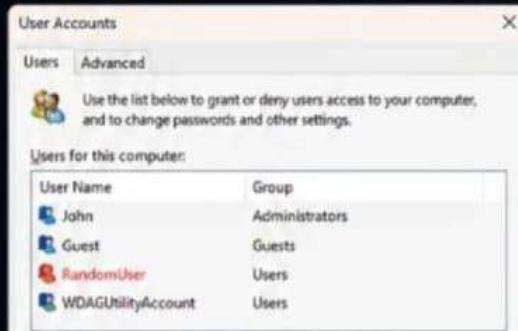
2 Type **netplwiz** and press Enter.



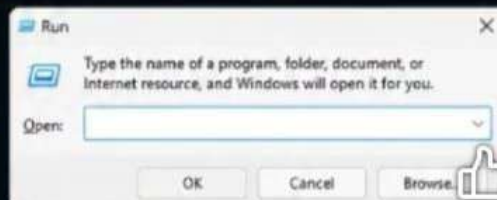
3 You'll see all the **user accounts** on your PC.



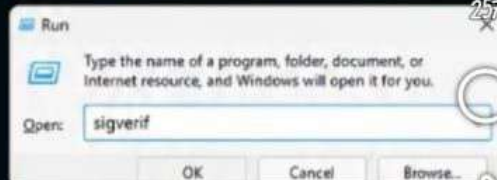
If you notice an account you don't recognize, **remove** it and click **OK**.



4 Now press **Windows + R** again.



5 Type **sigverif** and press Enter.



6 Click **Start** and let Windows scan your system. It may take a few minutes.



High-Tech Pro
Giulio Cercato - Beautiful

Ikuti



"Someone's Secretly Using Your PC RIGHT NOW? Check This Hidden Windows

Tambahkan komentar...





VIRUS & MALWARE DETECTION METHODS

Multiple layers of protection to detect known and unknown threats



1 SIGNATURE-BASED DETECTION

- Compares files with a database of known malware signatures.
- Effective against known viruses.



2 HEURISTIC ANALYSIS

- Analyzes code and looks for suspicious patterns or characteristics.
- Detects new or modified malware.



3 BEHAVIOR-BASED DETECTION

- Monitors the behavior of programs and processes.
- Detects malicious activities like file encryption, registry changes, etc.



4 REAL-TIME PROTECTION

- Continuously scans files, applications and system activities in real-time.
- Blocks threats before they cause damage.



5 SANDBOXING

- Runs suspicious files in an isolated and controlled environment.
- Observes behavior to detect malicious actions safely.



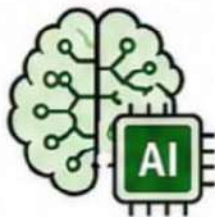
6 CLOUD-BASED DETECTION

- Sends file characteristics or telemetry to cloud servers.
- Leverages global threat intelligence to detect emerging threats.



7 MACHINE LEARNING / AI

- Uses machine learning models to identify malicious patterns.
- Detects previously unseen and zero-day malware.



8 ROOTKIT DETECTION

- Detects hidden malware, suspicious drivers, or unauthorized system modifications.
- Uncovers stealth threats deeply embedded in system.



9 NETWORK-BASED DETECTION

- Monitors network traffic for malicious activity.
- Detects command-and-control communications and suspicious connections.



10 INTEGRITY CHECKING

- Monitors critical system files and settings.
- Detects unauthorized changes or tampering.



KEY POINT

Modern antivirus solutions use multiple detection methods together (signatures + heuristics + behavior monitoring + cloud intelligence + machine learning) to provide stronger protection against both known and new malware.



OUTDATED DRIVERS: PROBLEMS & SOLUTIONS

Outdated drivers can cause many issues on your computer. Here are common problems and how to fix them.



1



SLOW COMPUTER PERFORMANCE

PROBLEM:

Graphics, storage, or chipset drivers may not work efficiently.

SOLUTION:

Update the affected driver through Windows Update or the manufacturer's official website.

2



DEVICE NOT WORKING

PROBLEM:

Wi-Fi, Bluetooth, USB, audio, or other hardware may stop working correctly.

SOLUTION:

Open Device Manager → right-click the device → Update driver.

3



SCREEN FLICKERING OR DISPLAY PROBLEMS

PROBLEM:

Old graphics drivers can cause flickering, low resolution, crashes, or poor gaming performance.

SOLUTION:

Install the latest compatible graphics driver from Intel, AMD, NVIDIA, or your PC manufacturer.

4



SYSTEM CRASHES OR BLUE SCREEN

PROBLEM:

Incompatible or outdated drivers can cause Windows crashes.

SOLUTION:

Update the problematic driver. If the issue started after an update, try Roll Back Driver.

5



NO SOUND OR AUDIO PROBLEMS

PROBLEM:

Outdated audio drivers can cause missing sound, crackling, or microphone issues.

SOLUTION:

Update or reinstall the audio driver.

6



NETWORK PROBLEMS

PROBLEM:

Old Wi-Fi or Ethernet drivers can cause connection drops or poor network performance.

SOLUTION:

Update the network adapter driver and restart the PC.

7



USB DEVICES NOT RECOGNIZED

PROBLEM:

Outdated chipset or USB controller drivers can prevent USB devices from working properly.

SOLUTION:

Update USB/chipset drivers and restart the computer.

8



GAMES RUNNING POORLY

PROBLEM:

Old GPU drivers may cause low FPS, crashes, or compatibility problems.

SOLUTION:

Install the latest stable graphics driver and keep Windows updated.

9



HARDWARE FEATURES MISSING

PROBLEM:

Some features may not work because the correct driver isn't installed.

SOLUTION:

Download the correct driver specifically for your PC model and Windows version.

10



DRIVER CONFLICTS

PROBLEM:

A new driver may conflict with an old or incompatible driver.

SOLUTION:

Remove the problematic driver, restart, and install a clean, compatible version.



IMPORTANT!

Don't download drivers from random websites. Use Windows Update, your PC manufacturer's support page, or the hardware manufacturer's official website.



Windows Update



PC Manufacturer Support Page



Hardware Manufacturer Official Website



ANIK

digital and
information technology



Learn • Fix • Automate

FOLLOW US



By HexSecTeam

Wi-Fi Network Scanner

Scan nearby Wi-Fi networks and view
SSID, signal, channel and security using Python

>_

Code:

```
# scan nearby Wi-Fi networks
import subprocess

# Run Windows command to get available networks
command = "netsh wlan show networks mode=bssid"
output = subprocess.check_output(command, shell=True)
decoded = output.decode(errors='ignore')

# Parse and display the results
networks = []
current = {}
for line in decoded.splitlines():
    line = line.strip()
    if line.startswith("SSID "):
        if current:
            networks.append(current)
            current = {}
        current["ssid"] = line.split(":", 1)[1].strip()
    elif "Signal" in line:
        current["signal"] = line.split(":", 1)[1].strip()
    elif "Channel" in line:
        current["channel"] = line.split(":", 1)[1].strip()
    elif "Authentication" in line:
        current["security"] = line.split(":", 1)[1].strip()
if current:
    networks.append(current)
print("\n[+] Nearby Wi-Fi Networks:\n")
for i, net in enumerate(networks, 1):
    print(f"[{i}] {net.get('ssid', 'N/A')} | "
          f"Signal: {net.get('signal', 'N/A')} | "
          f"Security: {net.get('security', 'N/A')} | "
          f"Channel: {net.get('channel', 'N/A')}")
```



python

SCANNING FOR WI-FI NETWORKS



>_

Output:

[1]	HomeLab_24		Signal: 92%		Security: WPA2-Personal		Channel: 6
[2]	CoffeeShop_FreeWiFi		Signal: 71%		Security: Open		Channel: 11
[3]	StudioMesh_5G		Signal: 84%		Security: WPA3-Personal		Channel: 149
[4]	GuestNet		Signal: 63%		Security: WPA2-Personal		Channel: 1
[5]	Office_AP		Signal: 56%		Security: WPA2-Enterprise		Channel: 44

 **Note:** Uses built-in Windows tool (netsh). No admin rights required.

Tips & Info

- ✓ Scans only nearby visible Wi-Fi networks.
- ✓ Best used on your own or authorized systems.
- ✓ Signal strength and channel may vary.
- ✓ Helpful for troubleshooting and awareness.



hexsecteam



hexsec_tools



hexsecteam

SECURE. AUTOMATE. EMPOWER.

TRANSFER LARGE FILES

USB FORMAT TRICK!



FAT32 ✗
MAX FILE SIZE
4GB

exFAT / NTFS ✓
MAX FILE SIZE
16EB+



**NO MORE
FILE SIZE LIMITS!**
FASTER. BIGGER. BETTER.

#SmartPCTricks

PERFECT FOR:

- 4K/8K VIDEOS
- GAME FILES
- BACKUPS
- BIG PROJECTS

HOW TO FORMAT YOUR USB (WINDOWS)

1 PLUG IN
YOUR USB



2 OPEN FILE
EXPLORER



3 RIGHT CLICK
USB DRIVE



4 CHOOSE FORMAT
exFAT or NTFS



FORMAT SMART. TRANSFER FAST.

**UNLOCK THE TRUE POWER
OF YOUR USB!**



RAM FAILURE

SYMPTOMS & TESTING METHODS

RAM (Random Access Memory) issues can cause system instability, data errors, and performance problems.



RAM FAILURE SYMPTOMS



1. FREQUENT BLUE SCREEN (BSOD)

Random system crashes with memory-related stop codes like `MEMORY_MANAGEMENT`, `KERNEL_DATA_INPAGE_ERROR`, etc.



2. SYSTEM FREEZES / HANGS

PC freezes or becomes unresponsive during normal use.



3. RANDOM RESTARTS

System restarts unexpectedly without any warning.



4. POOR PERFORMANCE

Slow performance, even with basic tasks or when opening applications.



5. FILE CORRUPTION

Files become corrupted or won't open properly.



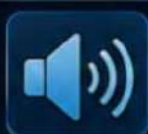
6. APPLICATION CRASHES

Programs crash frequently or show memory-related errors.



7. FAILURE TO BOOT

System fails to POST or won't boot at all.



8. BEEP CODES

Motherboard emits memory-related beep codes during startup.

TESTING METHODS



1. WINDOWS MEMORY DIAGNOSTIC

- Press Win + R, type `mdsched.exe`
- Choose "Restart now and check" for problems (recommended)
- System will restart and test RAM
- Check results after login



3. CHECK RAM SLOTS & MODULES

- Power off and unplug the PC
- Reseat the RAM modules
- Try one module at a time
- Test in different slots



5. CHECK EVENT VIEWER

- Open Event Viewer
- Go to Windows Logs > System
- Look for memory-related errors or warnings



MemTest86

2. MEMTEST86 (RECOMMENDED)

- Download MemTest86
- Create a bootable USB
- Boot from USB and let the test run (at least 2-4 passes)
- Any RED error = RAM problem



4. USE SYSTEM TOOLS / CMD

- Run: `sfc /scannow`
- Run: `DISM /Online /Cleanup-Image /RestoreHealth`
- Helps fix corrupted system files caused by bad RAM



6. BIOS / UEFI DIAGNOSTICS

- Some motherboards have built-in memory tests in BIOS/UEFI
- Run diagnostics if available



TIPS

- Always test with known good RAM if possible.
- Ensure your RAM is compatible with your motherboard.
- Keep your system and drivers up to date.

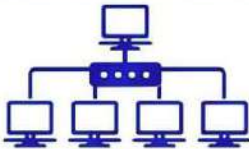


Early detection of RAM issues can prevent data loss and system damage.

SWITCH vs ROUTER vs FIREWALL

KEY DIFFERENCES AT A GLANCE

SWITCH



PRIMARY FUNCTION

Connects devices within the same network (LAN).



WORKS AT
OSI Layer 2
(Data Link Layer)



MAIN ROLE
Forwards data between devices using MAC addresses.



CONNECTS
Devices in the same LAN
(e.g., PC, Printer, IP Phone)



BROADCAST DOMAIN
Each port is a separate collision domain.
One broadcast domain (by default).

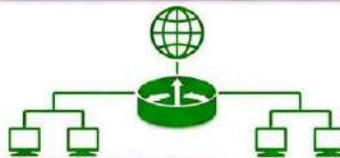


EXAMPLE USE
Connecting computers and other devices inside an office network.



IN SHORT
Switch connects devices within a network.

ROUTER



PRIMARY FUNCTION

Connects different networks together and routes data between them.



WORKS AT
OSI Layer 3
(Network Layer)



MAIN ROLE
Forwards data between networks using IP addresses.



CONNECTS
Different networks
(e.g., LAN to LAN, LAN to Internet)



BROADCAST DOMAIN
Separates broadcast domains.
Each interface is a separate broadcast domain.



EXAMPLE USE
Connecting your home/office network to the internet.



IN SHORT
Router connects networks and finds the best path for data.

FIREWALL



PRIMARY FUNCTION

Protects the network by monitoring and controlling incoming and outgoing traffic.



WORKS AT
OSI Layer 3, 4, and 7
(Network, Transport, Application Layer)



MAIN ROLE
Allows or blocks traffic based on security rules.



CONNECTS
Sits between trusted network (LAN) and untrusted network (e.g., Internet).



BROADCAST DOMAIN
Can separate and protect broadcast domains.



EXAMPLE USE
Protecting a network from unauthorized access and cyber threats.



IN SHORT
Firewall protects your network by controlling what traffic is allowed or blocked.



WINDOWS SYSTEM PASSWORD RECOVERY METHODS

Ways to Reset or Recover Your Windows Password



01 Microsoft Account Password Reset (Windows 10/11)



1. On the sign-in screen, click "I forgot my password".
2. You will be redirected to the Microsoft account recovery page.
3. Verify your identity using email, phone, or security questions.
4. Create a new password and sign in.

02 Password Reset Disk (Windows 10/11)



1. Insert the Password Reset Disk you created earlier.
2. On the sign-in screen, click "Reset password".
3. Follow the wizard to reset your password.

03 PIN Reset (Windows 10/11)



1. Click "I forgot my PIN" on the sign-in screen.
2. Verify your identity using your Microsoft account.
3. Set up a new PIN.

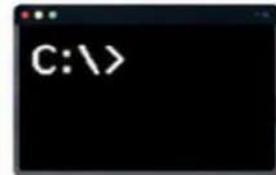
04 Local Account Password Reset (Security Questions)



1. On the sign-in screen, click "Reset password".
2. Answer your security questions.
3. Set a new password and sign in.

* Note: This option is available only if you set up security questions previously.

05 Command Prompt (Method)



1. Boot to Windows Recovery Environment (WinRE).
2. Open Command Prompt.
3. Use commands to reset the password (e.g., copy cmd.exe, replace utilman.exe).
4. Restart and reset password from sign-in screen.

06 Safe Mode (Method)



1. Boot into Safe Mode.
2. Sign in with another admin account.
3. Go to Control Panel > User Accounts > Manage another account.
4. Select the locked account and change the password.

07 Use Another Admin Account



1. Sign in with another admin account.
2. Open Settings > Accounts > Family & other users.
3. Select the user account.
4. Click "Change account type" or "Change password".

08 Windows Installation Media (Offline Reset)



1. Boot from Windows installation USB/DVD.
2. Open Command Prompt from setup.
3. Use commands to replace utilman.exe with cmd.exe.
4. Restart and reset password from sign-in screen.

09 Third-Party Password Reset Tools



1. Use trusted tools like:
 - Offline NT Password & Registry Editor
 - Lazesoft Recover My Password
 - PCUnlocker
2. Follow the tool instructions to reset the password.

* Note: Use trusted tools only.

10 Contact Support



1. Contact your system administrator or IT support.
2. For Microsoft Account issues, visit: account.live.com/acsrr
3. Provide required verification details to recover access.

PREVENT FUTURE PASSWORD ISSUES



Create a Password Reset Disk



Use a Microsoft Account



Enable PIN and Windows Hello



Keep Recovery Email and Phone Updated



Regularly Backup Important Data

IMPORTANT TIPS

- Keep your recovery options up to date.
- Do not share your password with anyone.
- Use strong and unique passwords.
- Enable 2-Factor Authentication wherever possible.



Windows 11

HOW TO ENTER SAFE MODE IN WINDOWS 11

STEP-BY-STEP GUIDE

Safe Mode helps you troubleshoot problems
and fix issues in Windows.

SAFE
MODE



FIX PROBLEMS

Resolve system
errors & crashes



TROUBLESHOOT

Fix driver, software,
or update issues



LIGHTWEIGHT

Loads only essential
system files



SAFE & SECURE

Better control while
troubleshooting

4 WAYS TO ENTER SAFE MODE IN WINDOWS 11

1 FROM SETTINGS (Works if Windows boots)

- 1 Press Windows + I to open Settings.
- 2 Go to System > Recovery.
- 3 Under Advanced startup, click Restart now.
- 4 After restart, follow Method 2 (Steps 5-7).



System > Recovery

Advanced startup

Restart your device to change startup settings,
including starting from a disc or USB drive

Restart now

2 USING ADVANCED STARTUP OPTIONS

- 1 Restart your PC.
- 2 As soon as the Windows logo appears, press and hold the Power button to force shut down.
- 3 Repeat this 2-3 times until you see "Preparing Automatic Repair".
- 4 Click Advanced options.
- 5 Click Troubleshoot.
- 6 Click Advanced options.
- 7 Click Startup Settings.
- 8 Click Restart.



Troubleshoot



Advanced
options



Startup
Settings

3 FROM STARTUP SETTINGS (AFTER RESTART)

- 1 After clicking Restart in Method 2, you will see a list of options.
- 2 Press 4 or F4 for Safe Mode.
- 3 Press 5 or F5 for Safe Mode with Networking.
- 4 Press 6 or F6 for Safe Mode with Command Prompt.
- 5 Your PC will restart and boot into Safe Mode.

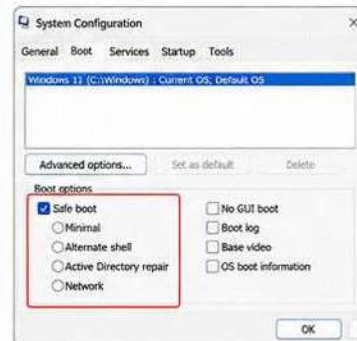
Startup Settings

Press a number to choose from the options below;
Use number keys or function keys F1-F9.

- 1) Enable debugging
 - 2) Enable boot logging
 - 3) Enable low resolution video
 - 4) Enable Safe Mode
 - 5) Enable Safe Mode with Networking
 - 6) Enable Safe Mode with Command Prompt
 - 7) Disable driver signature enforcement
 - 8) Disable early launch anti-malware protection
 - 9) Disable automatic restart after failure
- Press F10 for more options.
Press Enter to return to your operating system.

4 USING SYSTEM CONFIGURATION (MSCONFIG)

- 1 Press Windows + R, type msconfig and press Enter.
- 2 Go to Boot tab.
- 3 Check Safe boot and choose the type.
- 4 Click Apply > OK.
- 5 Restart your PC.



TYPES OF SAFE MODE



SAFE MODE

Starts Windows with
minimum drivers and
services.



SAFE MODE WITH NETWORKING

Same as Safe Mode,
but includes network
drivers and services.



SAFE MODE WITH COMMAND PROMPT

Opens Command Prompt
in Safe Mode for
advanced troubleshooting.

WHEN TO USE SAFE MODE?

- ✓ When Windows won't start normally
- ✓ After a problematic driver or update
- ✓ To remove malware or unwanted software
- ✓ To troubleshoot crashes or blue screen errors



QUICK TIPS

- ✓ Use Safe Mode to fix issues, then restart normally.
- ✓ If you can't boot at all, use a Windows installation USB to access Advanced Options.
- ✓ Some apps and features may not work in Safe Mode.

MORE TECH GUIDES & TIPS

techwalejha.com

Read full article link in the first comment

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- ✓ Share it with your friends
- ✓ Follow for more tech tips
- ✓ Stay updated!

⚠ **DISCLAIMER:** This guide is for educational purposes only. Use at your own risk.



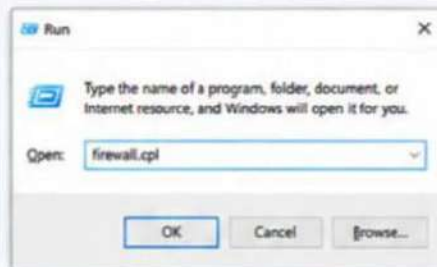
FIREWALL BLOCKING INTERNET – FIX

Follow these steps to fix internet issues caused by Windows Firewall.

✓ Try the steps in order for best results.

1 CHECK FIREWALL STATUS

- Press **Win + R**
- Type **firewall.cpl** and press Enter
- Click **Restore defaults** only if you are sure custom firewall rules are not needed.



2 ALLOW THE APP THROUGH FIREWALL

- Open Windows Security
→ Firewall & network protection
- Select **Allow an app through firewall**
- Enable the required application for Private and/or Public networks.

Allow apps to communicate through Windows Defender Firewall
To add, change, or remove allowed apps and ports, click Change settings.

What are the risks of allowing an app to communicate?

Change settings

Allowed apps and features:

Name	Private	Public
☒ Google Chrome	☒	☒
☒ Microsoft Edge	☒	☒
☒ Slack	☒	☒
☒ Remote Desktop	☒	☐

3 CHECK OUTBOUND RULES

- Press **Win + R** → type **wf.msc**
- Go to Outbound Rules
- Look for rules marked **Block**
- Disable only the rule that is blocking your application/network.

Windows Defender Firewall with Advanced Security

Windows Defender Firewall with
75

Inbound Rules

Outbound Rules

Connection Security Rules

Monitoring

Outbound Rules

Name	Group	Profile	Enabled	Action
Block Internet	-	All	Yes	Block
Block Chrome Update	-	All	Yes	Block
Block uTorrent	-	All	Yes	Block
Allow Internet	-	All	Yes	Allow
Allow Microsoft Edge	-	All	Yes	Allow

Actions

Outbound Rules

New Rule...

Filter by Profile

Filter by State

Filter by Group

View

Refresh

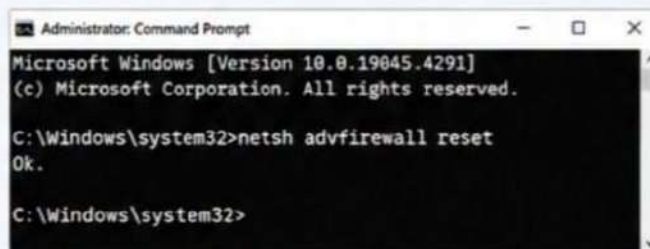
Help

4 RESET WINDOWS FIREWALL

Open Command Prompt as Administrator:

```
netsh advfirewall reset
```

Then restart the PC.



Firewall Reset
Successful

5 RESET NETWORK STACK

```
netsh winsock reset
netsh int ip reset
ipconfig /flushdns
ipconfig /release
ipconfig /renew
```



Then restart
Windows
afterward.



This helps fix:

- ✓ Connection issues
- ✓ DNS problems
- ✓ Network adapter errors



IMPORTANT: Don't permanently disable the firewall just to restore internet.
Find the blocking rule first—especially on an office/company PC
where firewall policies may be centrally managed.



FIX HARDDISK ISSUE

IN **30**
SECONDS



EASY BUILT-IN TOOL FOR WINDOWS PC!

1



Press **Windows Key** and type **Notepad**

2



Open **Notepad** and copy this script

```
@echo off  
echo y | chkdsk.exe c: /f /r  
shutdown.exe /r /t 10  
pause
```

3

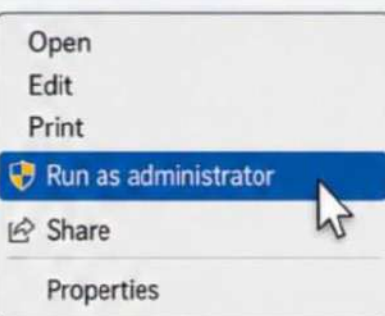


Click **File > Save As**
Name it

Check Disk Tool.bat
and save as **All Files**



Check Disk
Tool.bat



4



Right click on the file and **Run as administrator** to launch it

5



Your PC will **restart** and the disk check process will begin

6



Let the process **complete**
This will find and fix file system errors and bad sectors.

Checking disk...

Windows is checking the file system and scanning for problems...

This might take a while.

Progress: [██████████░░░░░░░░] 45%



This tool will fix many **hard disk issues** and make your PC **faster & smoother!**



SAVE THIS POST
FOR LATER!



FOLLOW & SUBSCRIBE
FOR MORE COMPUTER TIPS!

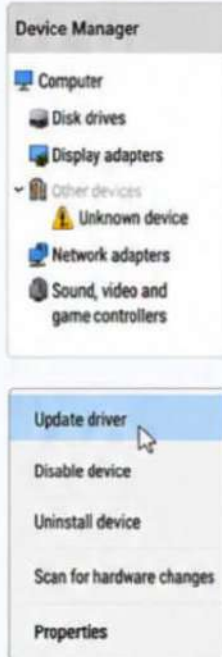


HOW TO INSTALL MISSING DRIVERS IN WINDOWS



METHOD 1: USE DEVICE MANAGER

- 1 Press Windows + X
- 2 Select Device Manager
- 3 Look for devices with:
 - Yellow warning mark
 - "Unknown Device"
- 4 Right-click the device
- 5 Select Update Driver
- 6 Choose:
Search automatically for drivers
- 7 Restart your PC



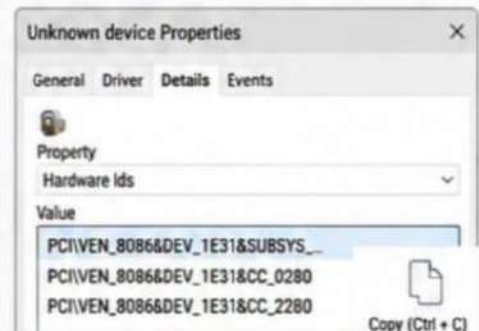
METHOD 2: INSTALL THROUGH WINDOWS UPDATE

- 1 Open Settings
- 2 Go to Windows Update
- 3 Click Advanced options
- 4 Select Optional updates
- 5 Open Driver updates
- 6 Select required drivers
→ **Download & Install**



METHOD 3: FIND DRIVER USING HARDWARE ID

- 1 Open Device Manager
- 2 Right-click the unknown device
- 3 Select Properties
- 4 Go to **Details** tab
- 5 Select **Hardware Ids**
- 6 Copy the ID (example: PCI\VEN_8086&DEV_....)
- 7 Search the hardware ID online or download from the manufacturer website



METHOD 4: INSTALL FROM MANUFACTURER WEBSITE

Download drivers from official websites:

- Laptop:** Dell / HP / Lenovo / ASUS / Acer support page
- Motherboard:** Intel / AMD / motherboard manufacturer
- Graphics:** NVIDIA / AMD / Intel

Install → Restart → Check Device Manager again.

METHOD 5: COMMAND LINE CHECK

Command Prompt (Admin)

```
C:\> pnputil /scan-devices
```

Scans for Plug and Play devices that do not have an installed driver.

To view installed drivers:

```
C:\> pnputil /enum-drivers
```

Displays all third-party drivers installed on your system.

COMMON MISSING DRIVERS

Device	Driver Needed
Network not working	Wi-Fi / LAN Driver
No sound	Audio Driver
Low display quality	Graphics Driver
USB not detecting	Chipset / USB Driver
Bluetooth missing	Bluetooth Driver



TIP: Avoid random "driver updater" software. They often install incorrect or unnecessary drivers. Use Windows Update or the device manufacturer's website whenever possible.



SSD NOT DETECTED IN BIOS

CAUSES & FIXES



If your SSD is not detected in BIOS, the computer is not seeing the drive at the hardware level. Try these fixes in order.



1 CHECK PHYSICAL CONNECTION

SATA SSD



- Shut down PC and remove power.
- Reseat the SATA data cable.
- Reseat the SATA power cable.
- Try another SATA port on the motherboard.
- Try another SATA cable.

M.2 NVMe SSD



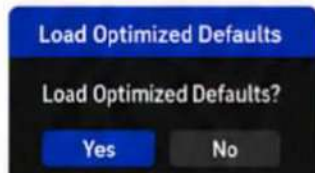
- Remove the M.2 SSD.
- Clean the connector gently.
- Reinstall it properly.
- Make sure the screw holds the SSD flat.
- Check if your motherboard supports that M.2 type (NVMe/SATA).

2 CHECK BIOS STORAGE SETTINGS



- Enter BIOS and check:
 - Storage Configuration
 - SATA Mode
- Use AHCI for normal SSD setups.
- Enable:
 - ✓ SATA Controller
 - ✓ NVMe Configuration (if available)

3 LOAD BIOS DEFAULT SETTINGS



- Enter BIOS
- Select Load Optimized Defaults
- Save & Restart
- Check SSD detection again

4 UPDATE BIOS/UEFI



- Older BIOS versions may not detect newer NVMe SSDs.
- Check motherboard/laptop manufacturer website.
- Update BIOS carefully.

5 TEST SSD IN ANOTHER SYSTEM



Another PC



USB to SATA Adapter
(for SATA SSD)



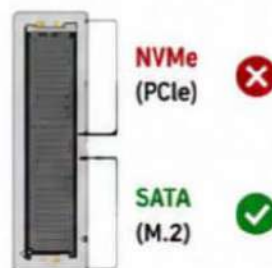
NVMe USB Enclosure
(for M.2 NVMe)



If another system also cannot detect it → SSD may have failed.

6 CHECK COMPATIBILITY

- ✓ M.2 SATA SSD in NVMe-only slot ✗
- ✓ NVMe SSD in M.2 SATA-only slot ✗
- ✓ PCIe generation limitations ✗
- ✓ Some M.2 slots share lanes with SATA ports ✗



7 IF BIOS DETECTS BUT WINDOWS DOESN'T



Disk Management

- ➔ Open Disk Management
- ➔ Initialize disk
- ➔ Assign drive letter
- ➔ Check drivers

QUICK DIAGNOSIS



NOT VISIBLE IN BIOS →
Cable / Slot / Compatibility / SSD Failure



VISIBLE IN BIOS BUT NOT WINDOWS →
Partition / Driver Issue





CARA INSTALL ULANG LAPTOP DENGAN BENAR



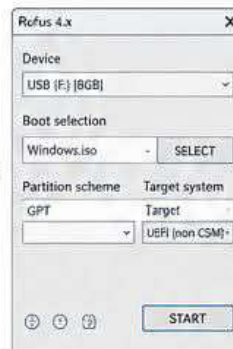
1 PERSIAPAN

- Backup data penting ke hardisk eksternal atau cloud.
- Siapkan Flashdisk minimal 8GB.
- Download File ISO Windows dari situs resmi Microsoft.
- Pastikan laptop terhubung ke charger.



2 BUAT BOOTABLE FLASHDISK

- Gunakan aplikasi Rufus untuk membuat bootable flashdisk.
- Masukkan flashdisk
- Pilih file ISO Windows
- Skema partisi: GPT
- Sistem target: UEFI
- Klik START



3 ATUR BOOT LAPTOP DARI FLASHDISK

- Colokkan flashdisk ke laptop.
- Nyalakan/restart laptop.
- Masuk ke menu Boot (tekan tombol: F2 / F12 / Esc / Del tergantung merk laptop).
- Pilih USB/Flashdisk sebagai boot device.



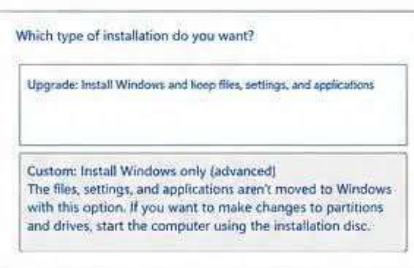
4 MULAI INSTALL WINDOWS

- Setelah boot dari flashdisk, akan muncul jendela instalasi Windows.
- Pilih Bahasa > Next
- Klik Install now



5 PILIH CUSTOM INSTALL (INSTALLASI BERSIH)

- Pilih: Custom: Install Windows only (advanced)
- Ini akan menghapus partisi lama dan menginstall Windows baru dari awal.



6 ATUR PARTISI

- Hapus semua partisi lama (kecuali System Reserved jika ada).
- Klik Drive 0 Unallocated Space > New untuk membuat partisi baru.
- Klik Next untuk mulai instalasi.



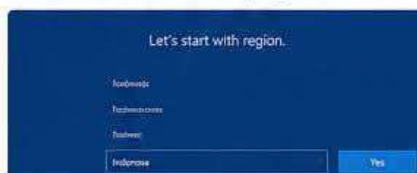
7 PROSES INSTALLASI

- Tunggu proses instalasi hingga selesai.
- Laptop akan beberapa kali restart otomatis.



8 SETUP WINDOWS

- Setelah instalasi selesai, lakukan pengaturan:
- Pilih wilayah
- Atur keyboard
- Hubungkan ke Wi-Fi
- Login dengan akun Microsoft
- Atur privasi
- Setelah selesai, Windows siap digunakan.



9 INSTALL DRIVER DAN APLIKASI

- Install driver dari website resmi laptop atau gunakan aplikasi bawaan (DriverPack Solution).
- Install aplikasi penting seperti:
 - Browser
 - Office
 - Antivirus
- Update Windows ke versi terbaru.



✓ TIPS PENTING

- Selalu backup data penting sebelum install ulang.
- Pastikan laptop terhubung ke charger selama proses instalasi.
- Jangan matikan laptop saat proses instalasi.
- Setelah selesai, selalu update Windows dan driver secara berkala.

⚠ PERINGATAN

Install ulang akan menghapus semua data di partisi sistem (C:). Pastikan data penting sudah dicadangkan.

TOMBOL MASUK BOOT MENU BEBERAPA MERK

- ASUS : F2 / Esc
- ACER : F2 / Del
- LENOVO : F1 / F2 / Novo Button
- HP : Esc / F9
- DELL : F2 / F12
- TOSHIBA : F12



★ **SELESAI!** Laptop Anda kini bersih dan siap digunakan seperti baru.



RECOVER DATA

from Corrupt Micro SD Card

USING USB DATA CABLE



SAFE & SECURE

Access your important data safely.



EASY CONNECTION

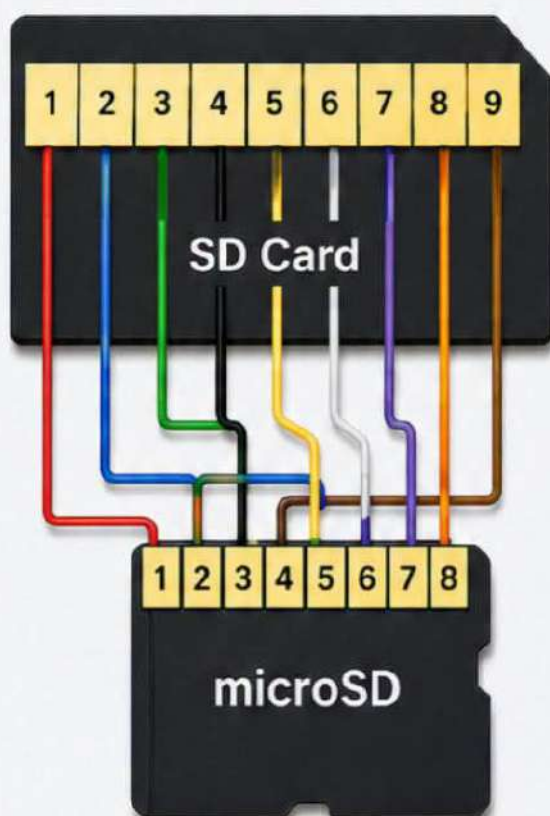
Simple USB data cable interface.



DATA RECOVERY

Recover photos, videos, documents & more.

WIRING DIAGRAM



This cable connects SD card pins to Micro SD card pins so the card can be accessed via USB for data recovery on your computer.



Note: This method is for educational purposes only. Use proper tools and handle with care.

SD CARD CONNECTION (9 PIN)

Pin	SD Signal	SPI Signal
1	CD/DAT3	CS
2	CMD	DI
3	VSS1	VSS1
4	VDD	VDD
5	CLK	SCLK
6	VSS2	VSS2
7	DAT0	DO
8	DAT1	X
9	DAT2	X

MICRO SD CONNECTION (8 PIN)

Pin	SD Signal	SPI Signal
1	DAT2	X
2	CD/DAT3	CS
3	CMD	DI
4	VDD	VDD
5	CLK	SCLK
6	VSS	VSS
7	DAT0	DO
8	DAT1	X



Tambah gambar di Excel semudah ini! Bisa logo, produk, foto, dan lainnya!



TUTORIAL CARA UPLOAD GAMBAR DI EXCEL

★ Mudah, cepat, rapi dan tidak berubah ukurannya!

Ikuti langkah mudah ini, hasilnya rapi dan profesional!



1 SIAPKAN FILE EXCEL

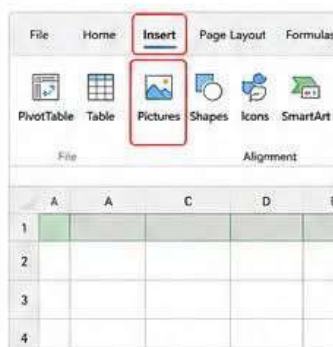
Buka file Excel tempat di mana Anda ingin menambahkan gambar.

	A	B	C	D
1	No	Nama Produk	Gambar	Harga
1	1	Laptop		8.000.000
2	2	Mouse		150.000
3	3	Keyboard		250.000
4	4	Headset		300.000
5	5	Flashdisk		100.000

Tips: Pastikan area/sel tempat gambar sudah disiapkan.

2 BUKA TAB INSERT

Klik tab Insert pada menu atas, lalu pilih Pictures.



Tips: Anda bisa pilih Pictures > This Device (untuk gambar dari komputer).

3 PILIH GAMBAR

Pilih gambar yang ingin diupload, lalu klik Insert.



Tips: Gunakan gambar dengan resolusi tinggi agar hasilnya jernih.

TIPS PENTING

- ✓ Gunakan format gambar JPG, PNG, atau GIF.
- ✓ Sesuaikan ukuran gambar agar proporsional.
- ✓ Gunakan fitur Move and Size untuk menempatkan gambar lebih presisi.
- ✓ Aktifkan fitur Lock Aspect Ratio saat mengubah ukuran agar tidak gepeng.
- ✓ Simpan file Excel setelah semua gambar dimasukkan.



4 GAMBAR BERHASIL DITAMBAHKAN

Gambar akan muncul di sel atau area yang dipilih.

No	Nama Produk	Gambar	Harga
1	Laptop		8.000.000
2	Mouse		150.000
3	Keyboard		250.000
4	Headset		300.000
5	Flashdisk		100.000

Tips: Gambar bisa digeser dan diatur ukurannya sesuai kebutuhan.

5 UBAH UKURAN GAMBAR

Klik gambar, lalu tarik sudutnya untuk mengubah ukuran.



Tips: Tahan Shift saat menarik sudut agar ukuran tetap proporsional.

6 ATUR POSISI GAMBAR

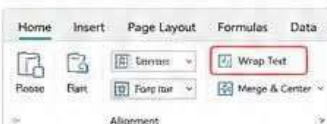
Seret gambar ke posisi yang diinginkan dalam sel atau area worksheet.

	A	C	D
1	No	Nama Produk	Harga
1	1	Laptop	8.000.000
2	2	Mouse	150.000
3	3	Keyboard	250.000
4	4	Headset	300.000
5	5	Flashdisk	100.000

Tips: Gunakan fitur Align pada tab Picture Format untuk posisi lebih rapi.

7 GUNAKAN FITUR WRAP TEXT

Agar teks dan gambar tertata rapi dalam sel.

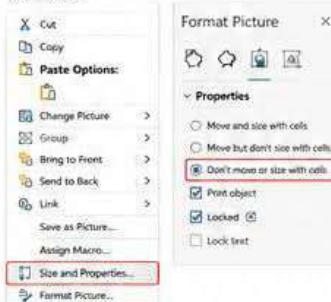


	A	C	D
No	Nama Produk	Gambar	Harga
1	Laptop		8.000.000

Tips: Aktifkan Wrap Text dan sesuaikan tinggi baris jika perlu.

8 KUNCI POSISI GAMBAR (OPSIONAL)

Klik kanan gambar > Size & Properties > Properties > pilih "Don't move or size with cells".



Tips: Gambar tidak akan bergeser saat data atau ukuran sel berubah.

9 HASIL AKHIR

Gambar terupload dengan rapi dan siap digunakan dalam laporan Anda!

No	Nama Produk	Gambar	Harga
1	Laptop		8.000.000
2	Mouse		150.000
3	Keyboard		250.000
4	Headset		300.000
5	Flashdisk		100.000

Tips: Simpan file Anda agar semua gambar tetap tersimpan.

★ MANFAAT

- 📊 Laporan lebih menarik dan informatif.
- 🔍 Mempermudah identifikasi data produk/foto.
- 📈 Presentasi data lebih profesional.
- 📊 Mendukung visualisasi data dengan lebih baik.
- ⌚ Hemat waktu dalam penyusunan laporan.

📝 CATATAN

- Pastikan ukuran file Excel tidak terlalu besar.
- Gunakan gambar secukupnya agar performa Excel tetap optimal.
- Jika file akan dibagikan, pastikan semua gambar ikut tersimpan.
- Cek kembali posisi dan ukuran gambar sebelum dicetak.



RINGKASAN LANGKAH



Follow **Smart Office ID**
untuk tips & tutorial Microsoft Excel setiap hari!





CORRUPTED WINDOWS SYSTEM FILES

Causes, Fixes & Prevention Guide



COMMON CAUSES



Sudden power shutdown



Malware or virus infection



Failed Windows update



Hard disk/SSD errors



Driver conflicts



Improper software installation

FIX 1: RUN SYSTEM FILE CHECKER (SFC)

- 1 Open Command Prompt as Administrator
- 2 Type the command:
`sfc /scannow`
- 3 Press **Enter**
- 4 Wait until the scan completes
- 5 Restart your PC

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.22631.3593]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>sfc /scannow

Beginning system scan. This process will take some time.

Beginning verification phase of system scan.
Verification 100% complete.

Windows Resource Protection found corrupt files
and successfully repaired them.

C:\Windows\System32>
```



Result:

"Windows Resource Protection found corrupt files and successfully repaired them" → **Fixed**



"Could not repair some files" → Continue with DISM

FIX 2: REPAIR WINDOWS IMAGE USING DISM



Open Command Prompt (Admin) and run:

```
DISM /Online /Cleanup-Image /RestoreHealth
```

After completion, run:

```
sfc /scannow
```

Restart your computer.



FIX 3: CHECK DISK ERRORS



Run the following command:

```
chkdsk C: /f /r
```

Restart when prompted.

This will scan and fix disk errors and recover bad sectors if any.

FIX 4: WINDOWS UPDATE REPAIR



Go to:

Settings → Windows Update → Check for Updates

Install pending updates and restart your PC.



FIX 5: ADVANCED REPAIR OPTIONS



If Windows cannot boot:

Troubleshoot → Advanced Options → Startup Repair

or use:

- System Restore
- Reset this PC (Keep my files)

PREVENTION TIPS



Use UPS during power cuts



Keep Windows updated



Avoid forced shutdowns



Monitor SSD/HDD health



Keep regular backups



Healthy System, Happy You!

Regular maintenance keeps your Windows fast, stable & error-free. Stay safe, stay productive!

★ SAVE THIS GUIDE FOR FUTURE PC TROUBLESHOOTING!

Printer Shows Offline – Fix

Follow these steps to get your printer back online!



1 Check the Printer

Make sure it's powered on and connected to Wi-Fi or USB.



2 Restart the Printer

Turn it off, wait 30 seconds, then turn it back on.



3 Check the Connection

Ensure your PC and Wi-Fi printer are on the same network.



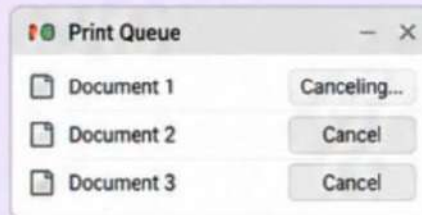
4 Disable Offline Mode

Go to Settings → Bluetooth & devices → Printers & scanners → Your printer → Open print queue → **Printer** → uncheck "Use Printer Offline."

☐ Use Printer Offline

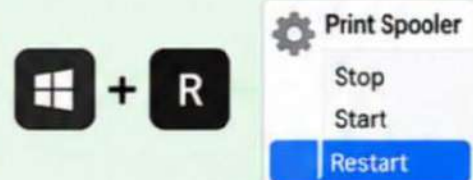
5 Clear Stuck Print Jobs

Cancel all pending documents in the print queue.



6 Restart Print Spooler

Press Win + R → type `services.msc` → find Print Spooler → Restart.



7 Remove & Re-add the Printer

Remove the printer from **Printers & scanners**, then add it again.



8 Update the Printer Driver

Install the latest driver from the printer manufacturer.



9 Restart Your PC

A simple reboot can refresh the printer connection.



10 Run Windows Printer Troubleshooter

Go to Settings → System → Troubleshoot → Other troubleshooters → Printer → Run.

Printer



Quick Tip: If a Wi-Fi printer keeps going offline, check whether its IP address has changed. Re-adding the printer using the correct IP can often fix the problem.

Small Fixes. A Smoother Tomorrow!

TechWhizAjith

TECH TIPS FOR A BETTER TOMORROW



Like



Share



Follow



Support

DAMAGED BOOT CONFIGURATION (BCD) CAUSES, ERRORS & EASY FIXES

WHAT IS BCD?

BCD (Boot Configuration Data) stores important information Windows needs to start:

- ✓ Windows boot location
- ✓ Operating system entries
- ✓ Boot settings
- ✓ Recovery options



ERROR MESSAGES YOU MAY SEE

- ! "The Boot Configuration Data file is missing"
- ! "0xc000000f – Boot Configuration Data error"
- ! "Windows failed to start"
- ! "Bootmgr is missing"



COMMON CAUSES



Sudden power failure



Failed Windows update



Corrupted system files



Hard disk/SSD errors



Incorrect partition changes



Malware or improper shutdown

1 AUTOMATIC STARTUP REPAIR



1. Boot from Windows Installation USB
2. Select Repair your computer
3. Go to:
Troubleshoot → Advanced Options → Startup Repair
4. Restart the PC

2 REBUILD BCD USING COMMAND PROMPT



Open Command Prompt from Advanced Options and run:

```
bootrec /scanos
```

Find Windows installations.

```
bootrec /rebuildbcd
```

Rebuild BCD.

If problems continue:

```
bootrec /fixmbr
```

Fix Master Boot Record.

```
bootrec /fixboot
```

Write a new boot sector.

Restart the computer.

3 RENAME AND RECREATE BCD



Open Command Prompt and run:

```
bcdedit /export C:\BCD_Backup
```

Backup current BCD.

```
attrib c:\boot\bcd -h -r -s
```

Remove hidden, read-only, and system attributes.

```
ren c:\boot\bcd bcd.old
```

Rename the old BCD file.

```
bootrec /rebuildbcd
```

Rebuild BCD.

4 CHECK DISK ERRORS



Open Command Prompt and run:

```
chkdsk C: /f /r
```

This checks and repairs disk errors and bad sectors.

PREVENTION TIPS



Keep Windows updates complete



Avoid forced shutdowns



Monitor SSD/HDD health



Keep a Windows recovery USB



Maintain regular backups

NOTE: If none of these methods work, your system may have severe disk or hardware issues. Consider professional help and data backup.





Printer Queue Stuck – Clear It

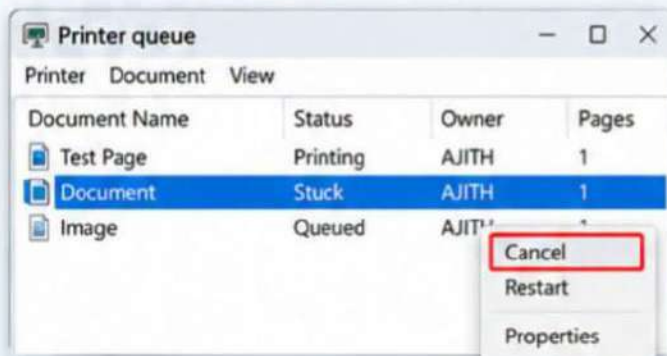
Print jobs stuck in the queue? No worries! Follow these simple steps to clear it and get your printer working again.

*Quick Fix
Easy Steps!*

Print Hassle Free Again! 😊

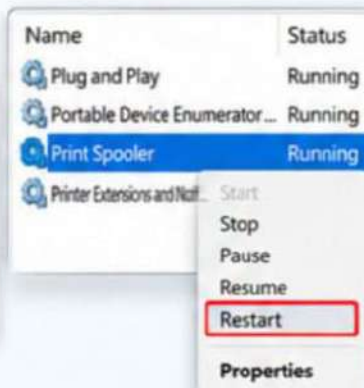
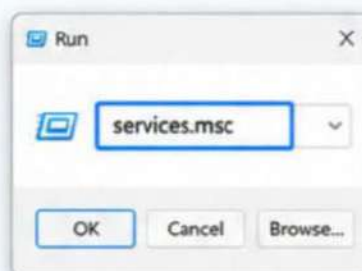
1 Cancel all print jobs

- Settings → Bluetooth & devices →
- Printers & scanners
- Select your printer → **Open print queue**
- **Cancel all documents.**



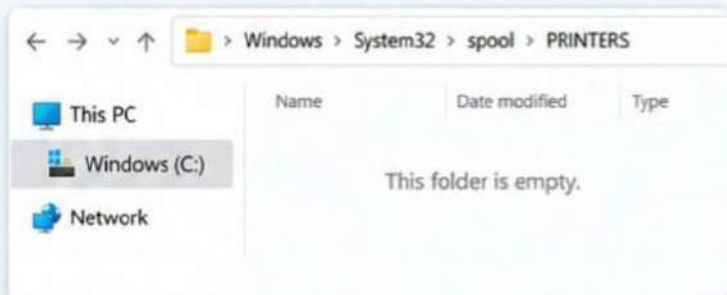
2 Restart Print Spooler

- Press **Win + R**
- Type **services.msc** and press **Enter**
- Find **Print Spooler**
- Right-click → **Restart**



3 Clear the spool folder (if still stuck)

- Stop **Print Spooler**
- Open this folder:
C:\Windows\System32\spool\PRINTERS
- Delete all files inside the folder.
- Start Print Spooler again.



4 Try printing again

- Open your document
- Click **Print** and check if it works.



*All Set!
Happy Printing!* 😊



Tip: Don't delete the **PRINTERS** folder itself—only its contents.

Simple Fixes for a Smoother Tech Life!

TechWhizAjith



*Follow
for more tech tips!*

How to Reinstall Windows 11

Step-by-Step Guide

Fix slow boot • Boost performance • Get a fresh start



Before You Start Important!

Since your PC takes about 45 minutes to load Windows, check if the drive (HDD/SSD) is failing. If it is, a clean installation may not fix the problem.

Back up your files and be sure you have everything you need.



What You Need



A working Windows PC
(to create the installer)



8 GB or larger USB flash drive
(Will be erased)



Internet connection



Your important files backed up
(to another drive/cloud)



Your Windows edition
(Home or Pro)



Keep your PC connected to power

1 Back Up Your Files

Before reinstalling, copy your important data:



- ✓ Desktop
- ✓ Documents
- ✓ Downloads
- ✓ Pictures / Videos
- ✓ Browser bookmarks & passwords (if not synced)
- ✓ Work files
- ✓ Software installers / license info



Do not proceed to the disk-partition stage until your data is safely backed up.

2 Create the Windows 11 USB

On another working PC, go to Microsoft's official page and follow these steps:

- 1 Click **Download Now**.
- 2 Run the **Media Creation Tool**.
- 3 Accept the license.
- 4 Choose **Create installation media for another PC**.
- 5 Select your language.
- 6 Choose **Windows 11**.
- 7 Select **64-bit**.
- 8 Choose **USB flash drive**.
- 9 Pick your USB device.
- 10 Let the tool create the bootable USB.



3 Boot the PC from USB

- 1 Insert the Windows USB.
- 2 Shut down the PC completely.
- 3 Turn it on and immediately press the **Boot Menu** key repeatedly.

Common keys: F12, F11, F9, Esc or F8 (depends on brand).

- 4 Select the **UEFI USB** option.
- 5 Windows Setup should appear.

Boot Menu

Windows Boot Manager
UEFI USB
EFI PXE Network

4 Start Windows Setup



- ✓ Choose your language, time/currency and keyboard.
- ✓ Click **Next**, then select **Install Windows 11**.

5 Product Key



If asked for a product key:

I don't have a product key

If your PC was already activated, activation will normally be restored after installation (if you install the same edition – Home or Pro).

6 Choose the Installation Type

For a true clean installation, select the option that will **delete everything**.

Where do you want to install Windows?

Name	Total size	Free space	Type
Drive 0 Partition 1	100 MB	100 MB	System
Drive 0 Partition 2	16 MB	16 MB	MSR (Reserved)
Drive 0 Partition 3	237.6 GB	237.6 GB	Primary
Disk 0 Unallocated Space	237.6 GB	237.6 GB	

- ✓ On Disk 0, delete all partitions until it becomes **Disk 0 — Unallocated Space**.
- ✓ Select the unallocated space and click **Next**.



Do NOT delete partitions on another disk containing your backup/files.

7 Let Windows Install

Installing Windows

- ✓ Copying files
- ✓ Installing features
- ✓ Installing updates
- ✓ Getting ready

This will take some time. Do not turn off your PC.



Windows will restart several times. If it keeps returning to Windows Setup, remove the USB or change the boot order back to the internal drive.

8 Complete Setup

- ✓ Connect to Wi-Fi / Ethernet.
- ✓ Sign in with your Microsoft account (if applicable).
- ✓ Run **Settings → Windows Update → Check for updates**.
- ✓ Install manufacturer drivers.
- ✓ Restart your PC.
- ✓ Install your essential programs.
- 8 Restore your backed-up files.



You're all set!
Enjoy your fresh Windows 11.



Pro Tips & Extra Advice

- ✓ If you're using an old HDD, consider upgrading to an SSD. It can reduce boot time from 45 minutes to under 1 minute!
- ✓ After installation, run Windows Update and install the latest drivers.
- ✓ Keep your system clean, avoid unnecessary startup apps, and scan for malware regularly.
- ✓ If you're unsure about your PC model or which boot key to use, tell me your brand (Dell, HP, Lenovo, etc.) and I'll give you the exact steps for your device.

C: DRIVE SHOWING RED INDICATOR?

Here's How to Fix It!

Free up space and keep your PC running smoothly.



Local Disk (C:)

12.3 GB free of 237 GB

This means your C: drive is running **low on free space**.

Let's fix it!

1 Check Storage Usage

Find out what is taking up space on your C: drive.



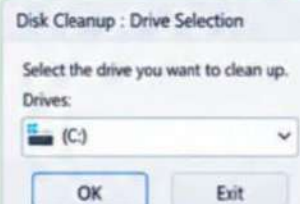
Open Settings
→ System
→ Storage



2 Run Disk Cleanup

Remove unnecessary system files.

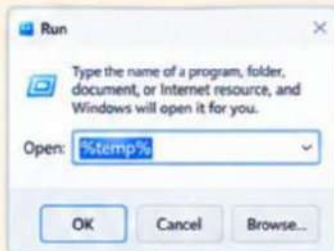
- 1 Press **Win + R**, type **cleanmgr** and press **Enter**.
- 2 Select **C:** drive.
- 3 Check the files and click **OK**.



3 Delete Temporary Files

Clear temporary files that are no longer needed.

- 1 Press **Win + R**, type **%temp%** and press **Enter**.
- 2 Delete all files you can.
- 3 Repeat the same with **temp**.



4 Empty Recycle Bin

Permanently remove deleted files.



5 Remove Unused Apps

Uninstall programs you no longer need.



Apps > Installed apps

Game XYZ	2.5 GB	Uninstall
Unused Software	1.8 GB	Uninstall
Old Tool	1.1 GB	Uninstall

- Go to Settings → Apps → Installed apps.
- Uninstall apps you don't use.

6 Enable Storage Sense

Let Windows automatically free up space.

System > Storage

Storage Sense

Automatically free up space, delete temporary files, and manage locally available cloud content.

☒ On

[Configure Storage Sense or run it now >](#)



Tips

- ✓ Keep at least **10-20%** of the C: drive free.
- ✓ Regularly run **Disk Cleanup**.
- ✓ Use **Storage Sense**.
- ✓ Remove old files and unused apps.
- ✓ Move large files (videos, photos) to another drive if possible.



Important

- ✗ Don't manually delete folders like C:\Windows, C:\Program Files, or unknown files from C:\Users.
- ✗ Only delete files you are sure about.

A Cleaner PC
A Smoother You! 😊

TechWhizAjith
Tips | Tricks | Troubleshooting

Keep Your PC
Healthy!



Remote Desktop Connection Failed

Unable to connect to the remote computer?
Follow these steps to fix the issue.



1



Enable Remote Desktop

Settings → System → Remote Desktop and turn it On.

2



Check the PC Name / IP Address

Press Win + R → type cmd
Run ipconfig and use the correct IPv4 address.

3



Check Network Connectivity

Open Command Prompt and run:
ping <IP-address>
If ping fails, check the network connection, VPN, firewall or routing.

4



Make Sure Remote Desktop Services is Running

Press Win + R → type services.msc
Find Remote Desktop Services and set it to Running.

5



Allow RDP through Firewall

Windows Security → Firewall & network protection → Allow an app through firewall.
Ensure Remote Desktop is allowed.

6



Check the RDP Port

Default RDP port: TCP 3389
Make sure it isn't blocked by the firewall or router.

7



Check the User Account

The account must have permission to connect through Remote Desktop.

8



Check Windows Edition

Windows Home editions cannot act as an RDP host using Microsoft's built-in Remote Desktop.



Pro Tip: If you are connecting from outside your office network, use a **VPN** for a secure connection instead of exposing port 3389 directly to the internet.

*Stay Connected
Work Smarter!*



How to Reset Windows 11

Step-by-Step Guide

Resetting Windows 11 can fix system issues, improve performance and give your PC a **fresh start**.



1 Open Settings



Press
Win + I
to open **Settings**.

2 Go to System



In Settings, click on **System** from the left sidebar.

3 Open Recovery



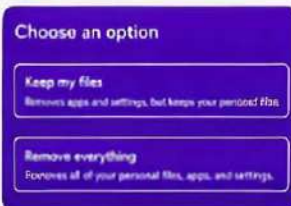
Scroll down and click on **Recovery**.

4 Click Reset PC



Under Recovery options, click **Reset PC**.

5 Choose an Option



Select one option:

- **Keep my files** (Recommended)
Removes apps and settings, but keeps your personal files.
- **Remove everything** (Full reset)
Removes all of your personal files, apps, and settings.

6 Additional Settings



Click **Change settings** to customize your reset preferences (optional).

7 Confirm Your Choice



Review the information and click **Reset** to continue.

8 Wait for the Process



Windows will reset your PC. This may take some time. Please wait...

9 PC Will Restart



Your PC will restart several times and **Windows 11** will be reset. **All done!** ✓



Back up your important files before resetting.



Ensure your PC is plugged in during the process.



Have your Windows product key ready.



The process may take 20-60 minutes depending on your PC.



Your PC will be like new, with better speed and stability!

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WHICH STEP WAS MOST HELPFUL FOR YOU?
COMMENT BELOW! 💬



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